

Rounding Almost Commuting Hamiltonians

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Commuting Hamiltonians lie at the boundary between classical constraint satisfaction and quantum many-body physics, exhibiting rich quantum structure while remaining more tractable than general noncommuting models. In contrast, physical Hamiltonians are rarely exactly commuting, which motivates the study of *almost commuting* Hamiltonians. Despite their relevance, the implications of approximate commutation are only poorly understood.

In this work, we show how to efficiently approximate any almost commuting 2-local qubit Hamiltonian by a commuting one: we give a new locality-preserving *algorithmic rounding technique* that maps any 2-local Hamiltonian $H = \sum_{i=1}^m h_i$ with $\|[h_i, h_j]\| \leq \varepsilon$ to a nearby Hamiltonian $\hat{H}$ whose terms pair-wise commute, and which is within overall distance $\|H - \hat{H}\| = O(m\varepsilon^{1/3})$.

As a consequence, we show that δ -approximations to the ground energy for ε -almost commuting 2-local qubit Hamiltonians lie in NP when $\delta \gg m\varepsilon^{1/3}$, extending the classical containment well beyond the commuting setting. Finally, we present two applications of our rounding framework: Gibbs sampling and fast Hamiltonian simulation for almost commuting systems.

1 Introduction

Commuting local Hamiltonians have been studied [BV04, AE11, Sch11, AKV18, IJ23, BH25] across condensed matter physics, quantum information theory, and computational complexity. From the perspective of constraint satisfaction, commuting local Hamiltonians sit at a boundary between classical and quantum problems. When all local terms commute, the ground-state problem reduces to the simultaneous satisfaction of a family of local constraints [BV04], analogous to a classical constraint satisfaction problem (CSP). Indeed, many commuting Hamiltonians admit ground states that can be described efficiently in *classical* terms, such as product states, stabilizer states, or tensor network states of bounded bond dimension [GHLS15]. The computational complexity of commuting Hamiltonians is thus significantly more tractable than that of general noncommuting models, with many cases lying in the classical complexity class NP [BV04, IJ23, BH25]. In condensed matter physics, commuting Hamiltonians serve as exactly solvable models that capture the essential structure of entire phases of matter. Prominent examples include stabilizer Hamiltonians for quantum error-correcting codes (e.g., the toric code [Kit03]),

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commuting-projector models for topological order [LW05], and fracton models with constrained dynamics [VHF16]. Formally, the k -local commuting Hamiltonian problem consists of Hamiltonians of the form

$$H = \sum_{i=1}^m h_i \quad \text{such that} \quad [h_i, h_j] = 0, \quad \forall i, j = 1, \dots, m \quad (1)$$

where all of the terms pair-wise commute¹ and each term is of unit norm $\|h_i\| \leq 1$ acting on at most k qubits. Commuting Hamiltonians retain intrinsically quantum features that sharply distinguish them from classical CSPs: although all of the Hamiltonian terms $\{h_i\}_{i=1}^m$ can be simultaneously diagonalized, the common eigenbasis may not be a product basis, and the ground states may exhibit long-range entanglement and topological order, such as the toric code [Kit03]. From a complexity-theoretic perspective, this places commuting Hamiltonians in an intermediate regime between classical satisfiability and fully quantum local Hamiltonians, providing a test ground for difficult problems such as the quantum PCP conjecture [AE11, Has13], the NLTS conjecture [ABN23], Gibbs state preparation [KB16] and fast thermalization [CB23].

Almost commuting Hamiltonians. Despite their utility, commuting Hamiltonians represent a highly nongeneric limit. Physical systems are rarely exactly commuting, and even infinitesimal noncommuting perturbations can produce dramatic qualitative changes. This motivates the study of *almost commuting* Hamiltonians, in which local terms fail to commute but only do so weakly; for example, when the Hamiltonian terms pair-wise " ε -almost commute" in the sense that

$$\|[h_i, h_j]\| \leq \varepsilon, \quad \forall i, j = 1, \dots, m. \quad (2)$$

This class of Hamiltonians captures a much richer range of physical phenomena, including entanglement generation, slow dynamics, and emergent conservation laws [Sac11, MRS24]. A simple and illustrative example is the transverse-field Ising model in the weak-field regime, described by the 2-local qubit Hamiltonian

$$H = - \sum_{\langle i, j \rangle} Z_i Z_j - h \sum_i X_i, \quad (3)$$

with $h \ll 1$. At $h = 0$, the Hamiltonian consists of exactly commuting terms and reduces to a classical Ising model with product-state ground states. Turning on a *weak* transverse field introduces local noncommuting terms whose pairwise commutator norms $\|[Z_i Z_j, h X_k]\|$ scale with h for $i = k$ or $j = k$, resulting in an almost commuting Hamiltonian. Generally speaking, even mild noncommutativity can cause the underlying physics to change qualitatively: the ground state becomes entangled, quantum fluctuations lift degeneracies, and the system supports nontrivial correlations which are absent in the commuting case [MRS24]. This illustrates how almost commuting Hamiltonians interpolate between classical constraint satisfaction and genuinely quantum many-body behavior.

The weak transverse-field Ising model might suggest that any almost commuting Hamiltonian can always be identified as a simple perturbation of an underlying commuting model. However, this intuition can be misleading in general. While the perturbation theory [Kat66] perspective is useful in identifying approximately commuting structure in very special or integrable models, this relies on strong assumptions—such as a known commuting parent

¹The *commutator* of two operators A and B is defined as $[A, B] = AB - BA$.

Hamiltonian or small, easily identifiable model parameters—which may not be present in generic local Hamiltonians.

From a computational standpoint, almost commuting Hamiltonians occupy a compelling intermediate regime. At first sight, they appear no longer classically reducible; and yet they are far from fully quantum Hamiltonians. Understanding whether their ground states and low-energy properties retain some of the tractability of commuting models—or instead exhibit the full hardness of general quantum constraint satisfaction problems—remains an intriguing open question.

Making almost commuting matrices commute. A naive idea for handling almost commuting systems, in general, is to simply discard Hamiltonian terms until the remainder commute; for example, one could consider the anti-commutation graph whose vertices correspond to Hamiltonian terms and whose edges connect pairs that fail to commute, and then try to find a *maximum independent set* (or some efficient approximation) [BH92]. The resulting Hamiltonian, obtained by retaining only the corresponding subset of terms, is exactly commuting by construction. However, this approach does not exploit the *magnitude* of the noncommutativity: its error is determined solely by how many terms are removed. In the worst case, even when all ε -almost commute for a small $\varepsilon > 0$, the procedure may discard a constant fraction of the terms, leading to a vacuous approximation error that scales with m (the number of Hamiltonian terms) with no improvement as $\varepsilon \rightarrow 0$. In contrast, a meaningful rounding procedure for almost commuting Hamiltonians should be ε -sensitive, with an error that decreases as the commutators become smaller. This suggests that an ε -sensitive rounding procedure must do more than just *pruning* a noncommuting interaction graph: it must exploit the algebraic structure of the Hamiltonian terms themselves.

A more sophisticated angle of attack for dealing with a set of *almost commuting* matrices is to explicitly find nearby *commuting* matrices; for example, starting with a pair

$$A, B \in \mathbb{C}^{d \times d} \quad \text{with} \quad \|[A, B]\| \leq \varepsilon \quad (4)$$

one might try to "round" to $\hat{A}, \hat{B}$ such that $[\hat{A}, \hat{B}] = 0$ with $\|\hat{A} - A\| \leq \delta(\varepsilon)$ and $\|\hat{B} - B\| \leq \delta(\varepsilon)$, for some $\delta(\varepsilon)$ such that $\delta(\varepsilon) \rightarrow 0$ as $\varepsilon \rightarrow 0$. The mathematical problem of rounding almost commuting matrices has a long and subtle history. For pairs of Hermitian matrices, Lin's theorem [Lin97] guarantees that rounding is indeed possible, and quantitative bounds on $\delta(\varepsilon)$ were later given by Hastings [Has09, Her22]. Surprisingly, the same is not possible for *triplets* of Hermitian matrices, as shown by Davidson [Dav85] and also by Hastings and Loring [HL10] who gave further topological obstructions towards rounding in terms of the *Bott index*. These results suggest that large collections of almost commuting Hermitian matrices can, in general, not be rounded to nearby commuting matrices unless further structure is enforced, hindering a direct application to almost commuting Hamiltonians. Nevertheless, many physically relevant non-commuting local Hamiltonians are semi-classical, and may very well be close to a commuting Hamiltonian, as the weak transverse-field Ising model in Equation (3) suggests. This raises the question:

Is it possible to round generic almost commuting Hamiltonians to nearby commuting Hamiltonians?

Progress on this question could shed new light on the extent to which weak noncommutativity is essential for the observed physics, versus merely a perturbation of an underlying commuting structure. Moreover, from a complexity perspective, it would allow us to better understand the boundary between classical satisfiability and inherently quantum optimization problems.

1.1 Our contributions

In this work, we overcome many prior limitations on rounding approximately commuting Hermitian matrices which have so far hindered applications to many-body Hamiltonians. We develop the first *locality-preserving algorithmic rounding technique* that allows one to simultaneously round large collections of almost commuting Hermitian matrices—avoiding the regimes where general no-go results apply—by exploiting the special structure of physical many-body Hamiltonians:

- (*Locality*.) Interactions in physical many-body Hamiltonians tend to be *local* in the sense that they involve a constant number of particles at a time; consequently, each term in the Hamiltonian is actually a highly non-generic Hermitian operator: it only acts non-trivially on a few sub-systems at a time and is equal to the identity everywhere else.
- (*Dimensionality*.) Many-body Hamiltonians tend to involve particles whose Hilbert space carries a fixed local dimension. In the case of two-level qubit systems (or "spin systems"), such constrained dimensionality can give rise to very useful properties, such as the fact that commutation between Hermitian single-qubit operators becomes *transitive*: if A, B commute and B, C commute, then A and C must also commute, provided that B is not a multiple of the identity.² In higher dimensions, this property can fail due to *degenerate eigenspaces*.³

Our rounding technique exploits these physical constraints, and allows us to efficiently approximate any almost commuting 2-local Hamiltonian by a nearby commuting one, while preserving locality. This is captured by our main result, which we informally state below.

Informal Theorem 1.1 (Formally, Theorem 5.1). *Let $H = \sum_{i=1}^m h_i$ be an ε -almost commuting 2-local Hamiltonian on n qubits. Then, there exists a deterministic classical algorithm that, given as input a description of H and the bound ε , runs in linear time and outputs a nearby commuting 2-local Hamiltonian $\hat{H} = \sum_{i=1}^m \hat{h}_i$ such that $\|h_i - \hat{h}_i\| \leq O(\varepsilon^{1/3})$, for all $i \in [m]$, and which is within overall distance $\|\hat{H} - H\| \leq O(m\varepsilon^{1/3})$.*

To the best of our knowledge, this yields the first *algorithmic* rounding technique for almost commuting 2-local qubit Hamiltonians which produces an explicit, quantitative error bound that vanishes as $\varepsilon \rightarrow 0$, unlike naive methods that fail to take the magnitude of the noncommutativity into account⁴. As a direct consequence of our rounding result, we show the following:

Informal Theorem 1.2 (Formally, Corollary 6.2). *The ε -almost commuting 2-local qubit Hamiltonian problem with m terms and energy promise gap⁵ δ is contained in NP when $\delta \gg m\varepsilon^{1/3}$.*

This extends the classical NP containment result of Bravyi and Vyalyi [BV04] well beyond the commuting setting. In addition, the above theorem also implies a new no-go result in the context of quantum PCPs [AE11, Has13] akin to [Ara11]. Specifically,

²We prove this statement in Lemma 3.5.

³For example, suppose that the matrices A and C act differently inside the same degenerate subspace of the matrix B . Then, both matrices can commute with B but they need not commute with each other.

⁴For a detailed comparison with the *non-constructive* rounding in [Ara11], see Section 1.3.

⁵Here we mean the difference in energy thresholds for the YES and NO cases of the local Hamiltonian problem. Note that in the formal statement of Corollary 6.2 we use the *relative* promise gap γ , which is equal to the energy promise gap divided by m .

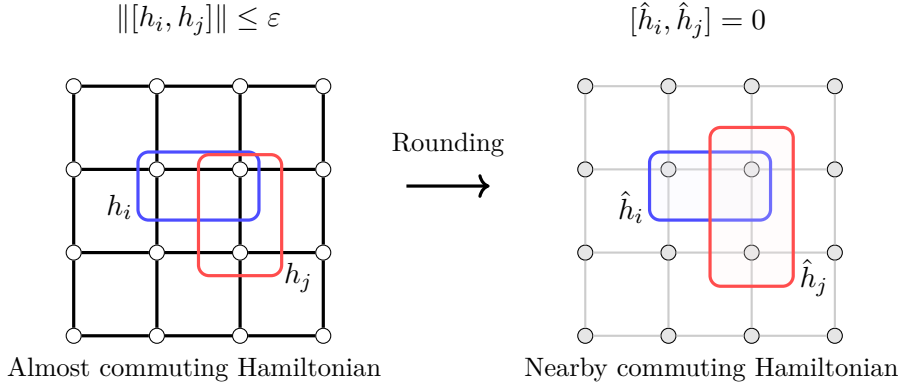


Figure 1: Illustration of our locality-preserving rounding transformation. Left: an almost commuting Hamiltonian $H = \sum_{i=1}^m h_i$ on a 2D lattice. Right: a nearby commuting Hamiltonian $\hat{H} = \sum_{i=1}^m \hat{h}_i$ of exactly the same locality. We remark that our rounding map does not require geometric locality.

it suggests that any qPCP-hard family of 2-local qubit Hamiltonians must exhibit some moderate amount of noncommutativity: assuming $\text{NP} \neq \text{QMA}$, any family of 2-local qubit Hamiltonians for which the γ -LH problem is QMA-hard for *constant* γ must allow for pairwise commutator norms of $\Omega(1)$.

More broadly, this places almost commuting Hamiltonians in an intermediate regime between classical constraint satisfaction and fully quantum optimization. It shows that, at least for 2-local qubit systems, genuine QMA-hardness requires a significant departure from the structure of commuting models, thereby linking a complexity-theoretic transition to a quantitative, operator-algebraic measure of noncommutativity.

Finally, in section A, we give a simple example Hamiltonian consisting of 2 terms that shows that a rounding error of $\Omega(\varepsilon^{1/2})$ is necessary. We discuss the prospects for making our bounds tight below in section 1.5.

1.2 Applications

As an application of our rounding technique for almost commuting quantum many-body systems, we present two algorithmic applications; the first one in the context of quantum Gibbs sampling and the second in the context of Hamiltonian simulation.

Quantum Gibbs sampling. The Gibbs ensemble connects microscopic descriptions of physical systems with their macroscopic thermodynamic behavior, allowing one to compute quantities such as free energies, effective partition functions, and equilibrium expectation values [PB11, Cha87, Cal85, LL80, Sac11, AS10, Mer65, Tuc10, LB14]. Given a local Hamiltonian H and an inverse temperature $\beta > 0$, the Gibbs state is given by

$$\rho_\beta(H) = \frac{e^{-\beta H}}{\text{Tr}[e^{-\beta H}]} \quad (5)$$

A recent line of work [TOV⁺11, CKF23, CKG23, BCL24, RW24] has proposed *quantum* algorithms for preparing such thermal states. Although these methods can offer theoretical speedups [BCL24, RW24], their performance typically hinges on either mixing-related characteristics or temperature-dependent quantities that are challenging to estimate or control in realistic settings.

A prominent category of Gibbs sampling tasks—one which typically admits polynomial-time algorithms with provable guarantees [HJ25, SLS⁺25]—is the *commuting case*, where

H is a *commuting* Hamiltonian. By leveraging the *Structure Lemma* [BV04], recent work [HJ25] showed that Gibbs sampling for large classes of commuting Hamiltonians even reduces to *Gibbs sampling* for certain *classical* Hamiltonians; these in turn can admit powerful classical methods such as Swendsen-Wang dynamics [SW87, FGW22]. While the commuting case has been widely studied, only little is known about the non-commuting case, let alone the *almost commuting* case.⁶ Currently, all existing quantum Gibbs sampling methods fail to take such almost commuting structure into account; in particular, the techniques for the commuting case are known to break down in the presence of (even mild) non-commuting interactions. This raises the question: *is it possible to extend quantum Gibbs sampling techniques for commuting Hamiltonians towards almost-commuting Hamiltonians?*

In Section 6.2, we give an affirmative answer to this question: we use our rounding technique to reduce Gibbs sampling for certain almost commuting 2-local Hamiltonians to Gibbs sampling for a nearby commuting Hamiltonian, thereby allowing one to once again make use of the full toolbox for the commuting case [HJ25, SLS⁺25]. Moreover, we show that a standard continuity bound on the stability of the Gibbs state suffices to ensure that the reduction is sound, and indeed approximately samples from the target state in Equation (5).

Faster Hamiltonian simulation. Simulating quantum-mechanical systems is one of the central tasks in quantum algorithms. It is well known that Trotter-based Hamiltonian simulation methods can perform better in the presence of commutativity in the Hamiltonian. In fact, for an exactly commuting Hamiltonian, the runtime of Hamiltonian simulation can be logarithmic in the simulation time t (as explained in Lemma 6.4), unlike the linear or worse scaling in the general case. This makes commutativity a useful algorithmic resource for real-time dynamics, not only a simplifying structural assumption for ground-state problems. For Hamiltonians that do not exactly commute, prior investigations [CST⁺21] have shown that commutator bounds can lead to faster Trotter-based simulation algorithms. Our setting is complementary: rather than analyzing the Trotter error directly, we first round the Hamiltonian to a nearby commuting one and then simulate the remaining discrepancy separately.

In Section 6.3, we observe that our rounding, together with the "interaction picture" method of Low and Wiebe [LW18], gives us a way to directly leverage the first-order commutators alone to speed up Hamiltonian simulation. In a nutshell, we apply the algorithm of [LW18] with the *fast* part of the Hamiltonian being the commuting rounding, and the *slow* part being the error term incurred in the rounding. Intuitively, the system is evolved mostly under a nearby exactly commuting Hamiltonian, while the genuinely non-commuting remainder is treated as a perturbation in the interaction picture. Apart from a commuting-simulation overhead and polylogarithmic factors, the time-dependent cost is governed by a norm bound for the latter term, controlled by the commutator error, rather than by the total norm of the Hamiltonian. In this way, the simulation cost is governed by how far the Hamiltonian is from the commuting regime, rather than by the full norm of the Hamiltonian itself.

⁶We remark that Schmidhuber et al. [SLS⁺25] recently studied another form of "almost commuting" Hamiltonians; namely, signed Pauli Hamiltonians with a *sparse* anti-commutation graph. In their work, they provide evidence that Gibbs sampling and optimization for such almost commuting systems is classically intractable, hinting at potential quantum advantage of the HDQI algorithm. However, their notion of almost commuting Hamiltonians is quite different from ours; in particular, our rounding algorithm does not seem to apply in their setting.

1.3 Related work

The work of [CRSV17] studied projections that pairwise nearly commute and used a predecessor (see Theorem 3.2 [CRSV17]) of the pinching technique we extend here. However, their techniques do not preserve the locality of the operators, and hence do not apply in our setting.

The specific question of rounding approximately commuting Hamiltonians to commuting Hamiltonians was first studied by Arad [Ara11]. In this work, it is shown that for two-local Hamiltonians on d -dimensional qudits on a D -regular interaction graph, whose terms are projectors, an "asymptotically good" rounding exists: that is, for fixed d and D , in the limit of the commutator error ε going to 0, the rounding error also goes to 0. The rounding error may depend on both d and D . However, the rounding is *non-constructive* and no explicit bound on the rounding error was established. Arad's techniques also do not easily lend themselves to algorithmic rounding, since they are based on a nonconstructive compactness argument inspired by a work of Halmos [Hal76]. The rounding results in [Ara11] are incomparable to ours: they are weaker in that they only apply to Hamiltonians whose terms are projectors; at the same time, they are also more general in that they apply to local dimensions greater than 2 as well. We remark that Arad's proof also makes use of local decompositions of the Hamiltonian terms, similar to the Pauli decompositions that we exploit. Comparing our results to his, we consider our main contribution to be the locality-preserving algorithmic rounding theorem with an explicit polynomial bound on the rounding error given by our analysis—to our knowledge, the first constructive rounding technique with an explicit error bound in terms of ε that goes to zero as $\varepsilon \rightarrow 0$. In fact, coming up with such a bound was listed by Arad as an open question, with the suggestion that it might follow easily from the techniques of [PS79]. This latter paper belongs to the rich literature on rounding approximately commuting high-dimensional matrices. We do not believe this literature is directly useful or relevant to our problem because we are interested specifically in roundings that preserve the *local structure* of the Hamiltonian.

Turning to complexity theory, our work connects to the extensive literature on commuting local Hamiltonians. Beginning with the result of Bravyi and Vyalı [BV04] that commuting 2-local qubit Hamiltonians lie in NP, subsequent works have extended NP containment results to broader commuting settings, including higher-locality qubit systems, certain qutrit systems, two-dimensional geometries, and more general interaction complexes [AE11, Sch11, AKV18, IJ23, BH25]. The central contribution of this work is to extend this viewpoint beyond exact commutativity. We show that sufficiently almost commuting 2-local qubit Hamiltonians can be rounded to nearby commuting Hamiltonians, thus extending the classical containment well beyond the commuting setting.

1.4 Technical Overview

Let us now proceed with a technical overview of the result in this paper. We assume familiarity with the basic formalism of quantum information, and refer the reader to Section 2 for a refresher.

To illustrate our rounding technique, we consider the simple (although already non-trivial⁷) example of an ε -almost-commuting 2-local Hamiltonian on a three-qubit triangle,

$$H = h_{1,2} + h_{1,3} + h_{2,3}, \quad (6)$$

⁷We remark that prior no-go results on rounding *general* Hermitian matrices already apply to triplets [Dav85].

where the i, j terms pair-wise ϵ -almost commute such that

$$\|[h_{1,2}, h_{1,3}]\| \leq \epsilon, \quad \|[h_{1,3}, h_{2,3}]\| \leq \epsilon \quad \text{and} \quad \|[h_{1,2}, h_{2,3}]\| \leq \epsilon \quad (7)$$

for some $\epsilon \in (0, 1)$. The goal is to identify an *exactly commuting* Hamiltonian $\hat{H} = \hat{h}_{1,2} + \hat{h}_{1,3} + \hat{h}_{2,3}$ with $[\hat{h}_{1,2}, \hat{h}_{1,3}] = [\hat{h}_{1,3}, \hat{h}_{2,3}] = [\hat{h}_{1,2}, \hat{h}_{2,3}] = 0$, which inherits the locality of the original Hamiltonian, and which is *nearby* in the sense that

$$\|h_{1,2} - \hat{h}_{1,2}\| \leq \delta(\epsilon), \quad \|h_{1,3} - \hat{h}_{1,3}\| \leq \delta(\epsilon), \quad \text{and} \quad \|h_{2,3} - \hat{h}_{2,3}\| \leq \delta(\epsilon) \quad (8)$$

for some function $\delta(\epsilon)$ with the property that $\delta(\epsilon) \rightarrow 0$ as $\epsilon \rightarrow 0$.

Let us first consider the Hamiltonian terms $h_{1,2}$ and $h_{1,3}$. Using a local expansion in terms of the Pauli basis $\{\sigma^{(i)}\}_{i=0}^3$ given in Lemma 4.1, we can expand the Hamiltonian terms in H as follows:⁸

$$h_{1,2} = \sum_{\alpha=0}^3 A_1^{(\alpha)} \otimes \sigma_2^{(\alpha)} \quad (9)$$

$$h_{1,3} = \sum_{\beta=0}^3 B_1^{(\beta)} \otimes \sigma_3^{(\beta)} \quad (10)$$

for some collections of Hermitian matrices $\{A_1^{(\alpha)}\}_\alpha$ and $\{B_1^{(\beta)}\}_\beta$ in $\mathbb{C}^{2 \times 2}$. Since we know that $\|[h_{1,2}, h_{1,3}]\| \leq \epsilon$, it is tempting to argue that the almost-commuting property must also "propagate" down to the collections of matrices $\{A^{(\alpha)}\}_\alpha$ and $\{B^{(\beta)}\}_\beta$. In Theorem 4.4, we show that this is indeed the case, leveraging the fact that $\{\sigma^{(i)}\}_{i=0}^3$ form an orthogonal basis of the space of 2×2 complex matrices under the Hilbert-Schmidt inner product. Concretely, we can argue that

$$\|[A_1^{(\alpha)}, B_1^{(\beta)}]\| \leq \epsilon, \quad \forall \alpha, \beta = 0, 1, 2, 3. \quad (11)$$

This observation allows us to effectively reduce the task of rounding a 2-qubit operator to that of rounding a collection of single-qubit operators; in other words, the goal now is to find *nearby* matrices $\{\hat{A}_1^{(\alpha)}\}_\alpha$ and $\{\hat{B}_1^{(\beta)}\}_\beta$ which perfectly pairwise commute i.e. $[\hat{A}_1^{(\alpha)}, \hat{B}_1^{(\beta)}] = 0$, for all α, β .

Pinching. How can we make two almost-commuting 2×2 Hermitian matrices commute? Here, we take inspiration from the well-known *pinching technique* for projectors (e.g. [CRSV17]). Suppose that $A, B \in \mathbb{C}^{2 \times 2}$ are arbitrary Hermitian matrices such that $\|[A, B]\| \leq \epsilon$ and $\Delta(A) > 0$; for example, as in (11) when A is not scalar. First, we observe that the spectral theorem allows us to expand the matrix A as

$$A = \lambda_{\min}(A) \Pi + \lambda_{\max}(A) (I - \Pi), \quad (12)$$

where $\lambda_{\min}, \lambda_{\max} \in \mathbb{R}$ are eigenvalues and Π and $(I - \Pi)$ are orthogonal projectors onto the eigenspaces. To *pinch* B by A (concretely, by the eigenspace projectors associated with A), we map

$$B \mapsto \hat{B} = \Pi B \Pi + (I - \Pi) B (I - \Pi). \quad (13)$$

⁸Here, the subscript indicates the qubit and we omit single-qubit identity operators for ease of notation.

Because $\hat{B}$ is now diagonal in the eigenbasis of A , we can see that $[\hat{B}, A] = 0$, and hence the two matrices perfectly commute. Moreover, a simple argument (formally proven in Lemma 3.2) reveals that the distance between $\hat{B}$ and B is at most

$$\|\hat{B} - B\| \leq \frac{\|[A, B]\|}{\Delta(A)} \leq \frac{\varepsilon}{\Delta(A)}, \quad (14)$$

where $\Delta(A) = |\lambda_{\max}(A) - \lambda_{\min}(A)|$ is the spectral gap of A . This immediately presents a problem: if the eigenvalues of A are extremely close to one another, the spectral gap $\Delta(A)$ can be vanishingly small, thereby causing the distance between $\hat{B}$ and B to blow up. In general, we may not have any control over the spectral properties that arise from the Pauli decompositions in (9) and (10).

Gap or snap? In order to explain how we get around the spectral gap issue, let us again revisit the example of two Hermitian matrices $A, B \in \mathbb{C}^{2 \times 2}$ with $\|[A, B]\| \leq \varepsilon$. We introduce a *threshold parameter* $\eta \in (0, 1)$ (to be determined later) and distinguish between the following cases:

- (the *gapped* case) $\Delta(A) \geq \eta$: in this case, the spectrum of A is sufficiently gapped, and the pinching bound in (14) translates into a stable upper bound of $\|\hat{B} - B\| \leq \frac{\varepsilon}{\eta}$.
- (the *nearly-degenerate* case) $\Delta(A) < \eta$: in this case, the pinching bound in (14) may blow up; fortunately, however, a small gap $\Delta(A)$ necessarily means that A must also be *close* to a multiple of the identity. Instead of pinching B , we can simply *snap* A towards $\hat{A} = \frac{\text{Tr}(A)}{2}I$. Importantly, this makes $\hat{A}$ trivially commute with not just B but in fact any other matrix as well—an important fact we will crucially exploit later on. A simple calculation (formally shown in Lemma 3.4) reveals that $\|\hat{A} - A\| = \Delta(A)/2$, and thus $\|\hat{A} - A\| < \eta$.

For example, letting $\eta = \sqrt{\varepsilon}$, we can always find a *nearby* commuting matrix within distance $\sqrt{\varepsilon}$; either by pinching B to get $\hat{B}$ such that $[\hat{B}, A] = 0$ and $\|\hat{B} - B\| \leq \frac{\varepsilon}{\eta} = \sqrt{\varepsilon}$, or by snapping A to get $\hat{A}$ such that $[B, \hat{A}] = 0$ and $\|\hat{A} - A\| < \eta = \sqrt{\varepsilon}$. Note that the 2×2 qubit structure in the "gap or snap" approach for $A, B \in \mathbb{C}^{2 \times 2}$ is crucial; indeed, in the qudit case, it is entirely possible for operators to have no spectral gap at all, and yet at the same time it is also not possible to simply "snap" them towards the identity.

Locality-preserving rounding. Equipped with the "gap or snap" paradigm, we explain a slightly simplified version of our full *locality-preserving rounding technique* on the ε -almost commuting Hamiltonian $H = h_{1,2} + h_{1,3} + h_{2,3}$ from before. (We will indicate the step that was simplified when we reach it in our explanation.) To this end, we once again use local expansions in terms of the Pauli basis $\{\sigma^{(i)}\}_{i=0}^3$. While our earlier expansions in (9) and (10) only involved one of the qubit systems, our full rounding technique makes use of Pauli expansions on both of the qubits; that is, for each of the terms h_{12}, h_{13} and h_{23} , we consider both possible expansions as follows:

$$h_{1,2} = \sum_{\alpha=0}^3 A_1^{(\alpha)} \otimes \sigma_2^{(\alpha)} = \sum_{\alpha=0}^3 \sigma_1^{(\alpha)} \otimes A_2^{(\alpha)} \quad \text{for} \quad \{A_1^{(\alpha)}\}_{\alpha=0}^3, \quad \{A_2^{(\alpha)}\}_{\alpha=0}^3 \quad (15)$$

$$h_{1,3} = \sum_{\beta=0}^3 B_1^{(\beta)} \otimes \sigma_3^{(\beta)} = \sum_{\beta=0}^3 \sigma_1^{(\beta)} \otimes B_3^{(\beta)} \quad \text{for} \quad \{B_1^{(\beta)}\}_{\beta=0}^3, \quad \{B_3^{(\beta)}\}_{\beta=0}^3 \quad (16)$$

$$h_{2,3} = \sum_{\gamma=0}^3 C_2^{(\gamma)} \otimes \sigma_3^{(\gamma)} = \sum_{\gamma=0}^3 \sigma_2^{(\gamma)} \otimes C_3^{(\gamma)} \quad \text{for} \quad \{C_2^{(\gamma)}\}_{\gamma=0}^3, \quad \{C_3^{(\gamma)}\}_{\gamma=0}^3 \quad (17)$$

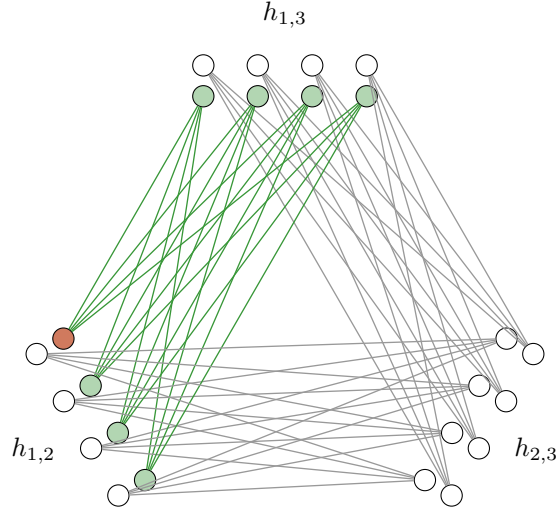


Figure 2: **The commutation graph of the 3-qubit triangle Hamiltonian.** The vertices represent matrices from the local Pauli expansion of each Hamiltonian term; the inner and outer rows show the expansions about qubits 1, 3 for $h_{1,3}$, 1, 2 for $h_{1,2}$, and 2, 3 for $h_{2,3}$, respectively. The edges represent near-commutation across Hamiltonian terms, as in (18), (19) and (20). For each qubit, our rounding procedure selects a *pivot* with a large spectral gap (e.g., the red vertex) and uses it to *pinch* all of the relevant vertices belonging to the same qubit (e.g. the green vertices). While vertices belonging to the same cluster are not required to commute, we also choose to *pinch* within clusters to exploit *transitivity*: if two green vertices each commute with the gapped pivot, they must also commute between each other. Pinching on the qubit shared between $h_{1,2}$ and $h_{1,3}$, the *pinching operator* $\mathcal{P}_1$ ensures that the pinched vertices all mutually commute (now, green edges); this implies that pinched Hamiltonian terms now exactly commute as $[\mathcal{P}_1(h_{1,2}), \mathcal{P}_1(h_{1,3})] = 0$.

For convenience' sake, we refer to the left decomposition in Equation (15) as the decomposition "about qubit 1", and the right decomposition in the same equation as the decomposition "about qubit 2"—and similarly for the others. Note that each expansion introduces a collection of four Hermitian matrices in $\mathbb{C}^{2 \times 2}$: in the decomposition "about qubit i ", these four matrices act on qubit i . In a slight abuse of notation, we use the same variable and rely on the subscript to indicate which qubit the matrices act on, as well as which of the two sets of matrices we are referring to. Thanks to our previous insight that the ε -almost-commuting property of $h_{1,2}$, $h_{1,3}$ and $h_{2,3}$ "propagates" down to the collections of matrices acting on the same qubit (formally, Theorem 4.4), we get

$$\|[A_1^{(\alpha)}, B_1^{(\beta)}]\| \leq \varepsilon, \quad \forall \alpha, \beta = 0, 1, 2, 3 \quad (18)$$

$$\|[A_2^{(\alpha)}, C_2^{(\gamma)}]\| \leq \varepsilon, \quad \forall \alpha, \gamma = 0, 1, 2, 3 \quad (19)$$

$$\|[B_3^{(\beta)}, C_3^{(\gamma)}]\| \leq \varepsilon, \quad \forall \beta, \gamma = 0, 1, 2, 3 \quad (20)$$

We may visualize the commutation structure of the triangle Hamiltonian as a graph $G = (V, E)$; here the vertex set V consists of the 24 matrices in (15), (16) (17), and E contains edges between vertex pairs which are ε -almost commuting according to (18), (19) and (20). We can group vertices belonging to the same Hamiltonian term together, and we can use a top and bottom vertex layer within each cluster to represent the two possible Pauli expansions (e.g., see Figure 2).

Choose $\eta = \varepsilon^{1/3}$. Then, our locality-preserving rounding takes place as follows:

1. (Partitioning:) Split the almost-commuting Hamiltonian H into a *gapped* and *nearly-degenerate* part (depending on η) such that $H = H_{\text{gap}} + H_{\text{deg}}$, where

- H_{gap} features Hamiltonian terms for which both sets of the two possible Pauli expansions contain a matrix of spectral gap of at least η (these particular matrices will enable us to round to nearby commuting matrices via *pinching*), and
 - H_{deg} features Hamiltonian terms that do not satisfy the gap condition: morally, these terms act "almost trivially" on at least one of the qubits they touch.
2. (Snap:) For each term $h_{i,j} \in H_{\text{deg}}$, we use snapping to round it to a term that is *exactly* trivial on at least one qubit. Concretely, suppose $h_{1,2}$ is in H_{deg} because its decomposition

$$h_{1,2} = \sum_{\alpha=0}^3 A_1^\alpha \otimes \sigma_2^{(\alpha)}$$

is not gapped—that is, each matrix A_1^α has spectral gap smaller than η . Then snapping simply consists of replacing each A_1^α in the decomposition with $\frac{\text{Tr}(A_1^\alpha)}{2} I_1$, yielding a new operator $h'_{1,2}$ that is effectively 1-local, and may in fact be simply a multiple of the identity. As a **simplifying assumption**, we will suppose that all the post-snapping operators $h'_{i,j}$ are in fact multiples of identity: the general case including 1-local operators is more cumbersome, requiring an additional round of snapping with the same gap parameter, but conceptually no more difficult than this special case. Let H' be H after snapping has been performed (with all the terms in H_{gap} unchanged). This step incurs an error of $O(\eta)$.

3. (Gap:) We are now in the position that for every Hamiltonian term that acts non-trivially on a qubit, at least one operator in its Pauli decomposition for that qubit is guaranteed to have a spectral gap. Intuitively, the spectral gap means these gapped operators have a "strong opinion" about the correct basis in which to pinch this qubit, and our algorithm will listen to them. More precisely, for any qubit i that is nontrivially acted on by at least two terms of H' , pick *any* gapped operator in the decomposition "about qubit i " of the terms acting on i , and set it to be the *pivot* R_i associated with i . (If a qubit does not have two nontrivial terms, set the pivot to be identity.) Once a pivot has been identified for each qubit, *pinch* each Hamiltonian term $h'_{i,j}$ simultaneously by the pivots R_i and R_j : mathematically,

$$h'_{i,j} \mapsto \hat{h}_{i,j} = (\mathcal{P}_i \otimes \mathcal{P}_j)(h'_{i,j}),$$

where $\mathcal{P}_i$ is the *pinching superoperator* given by

$$\mathcal{P}_i(X) = \Pi_i X \Pi_i + (I - \Pi_i) X (I - \Pi_i) \quad (21)$$

and where Π_i and $I - \Pi_i$ are the eigenspace projectors of the i -th pivot element; when $R_i = I$, we set $\mathcal{P}_i = \text{id}$. The Hamiltonian $\hat{H} = \sum_{(i,j)} \hat{h}_{i,j}$ is the final product of our rounding procedure.

After the pinching process is complete, it is easy to see that all the terms $\hat{h}_{i,j}$ are forced to commute with each other by construction. What is harder to see is that this does not incur a large error. This is essentially for two reasons:

- **Propagation:** If we imagine pinching $h'_{i,j}$ by a pivot arising from a *different* Hamiltonian term $h'_{i,k}$, then this pivot is guaranteed to almost commute—up to error ε —with all terms in the decomposition of $h'_{i,j}$ by the fact that $h'_{i,j}$ and $h'_{i,k}$ approximately commute, together with Theorem 4.4.

- **Transitivity:** If we imagine pinching $h'_{i,j}$ by a pivot arising from the *same* Hamiltonian term $h'_{i,j}$, then this pivot can be shown to almost commute with all other terms in the decomposition of $h'_{i,j}$ by using the *transitivity* of commutation for gapped matrices, by passing through another term $h'_{i,k}$. This is illustrated in Figure 2. The transitivity uses the gapped property of $h'_{i,k}$, and yields a slightly worse commutator error of $O(\varepsilon/\eta)$.

Thus, the pivot with which we pinch a given Hamiltonian term commutes with that term up to error at most $\kappa = O(\varepsilon/\eta)$, and it can be shown that the error induced in pinching is at most $O(\kappa/\eta) = O(\varepsilon/\eta^2)$. (We refer to Lemma 5.2 for a formal analysis of the pinching operation.) Our choice of $\eta = \varepsilon^{1/3}$ balances the errors from the gap and snap steps, giving us an overall error of $O(\varepsilon^{1/3})$. In the more general case, where our simplifying assumption in the snapping step is removed, it turns out a second round of snapping is needed: since snapping does not increase pairwise commutator norms, this second round uses the same gap parameter $\eta = \varepsilon^{1/3}$ and contributes $O(\varepsilon^{1/3})$ error, leaving the final bound $O(\varepsilon^{1/3})$ unchanged.

Notice that the Hamiltonian $\hat{H}$ inherits the same locality from the original Hamiltonian H ; this is due to the *local* nature of our pinching and snapping techniques. In Theorem 5.1, we show that our rounding technique—when applied to the triangle Hamiltonian—yields an *exactly commuting* Hamiltonian $\hat{H} = \hat{h}_{1,2} + \hat{h}_{2,3} + \hat{h}_{1,3}$ with $[\hat{h}_{1,2}, \hat{h}_{1,3}] = [\hat{h}_{1,3}, \hat{h}_{2,3}] = [\hat{h}_{1,2}, \hat{h}_{2,3}] = 0$ and

$$\|H - \hat{H}\| \leq O(\varepsilon^{1/3}).$$

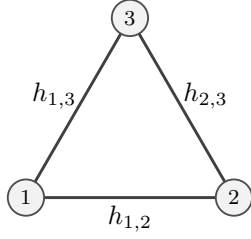
Our rounding technique thus rests on two important structural facts: (i) approximate commutation of Hamiltonian terms propagates to the single-qubit operators in their Pauli decompositions, and (ii) *pinching* against gapped local pivots enforces exact commutation while preserving locality. In this sense, our procedure is a consistent local change of basis across the interaction graph, with *snapping* handling terms which are too close to the identity to select a stable basis.

Concrete 3-qubit example. We now illustrate the rounding procedure on a concrete triangle Hamiltonian on three qubits, shown in Figure 3.

The terms $h_{1,2}$ and $h_{2,3}$ commute exactly: both are controlled on the shared qubit, qubit 2, in the standard basis. On the other hand, $h_{1,3}$ acts in the X basis on qubits 1 and 3, and therefore commutes only approximately with $h_{1,2}$ and $h_{2,3}$; the obstruction comes from the Z -basis components of $h_{1,2}$ and $h_{2,3}$, whose coefficients are suppressed by the factor $1/100$.

This example is useful because it rules out a naïve sequential approach to rounding. One might try first to round the already commuting pair $h_{1,2}, h_{2,3}$ to an exactly commuting pair $\hat{h}_{1,2}, \hat{h}_{2,3}$, without taking $h_{1,3}$ into account, and only afterwards try to round $h_{1,3}$ to some $\hat{h}_{1,3}$ that commutes with both of them. However, for the particular pair $h_{1,2}, h_{2,3}$ above, there is no operator on qubits 1 and 3 that acts nontrivially on both qubits and commutes exactly with both $h_{1,2}$ and $h_{2,3}$. Thus any successful rounding must modify at least one of $h_{1,2}$ or $h_{2,3}$, despite the fact that these two terms commute perfectly with each other. This illustrates why the rounding procedure must be global in nature: the choice of how to round one pair of terms cannot be made independently of the remaining interactions.

Let us now see how the one-shot rounding procedure handles this example. All three terms are gapped in the relevant sense in both of their possible Pauli decompositions, so



$$\begin{aligned}
h_{1,2} &= X \otimes |0\rangle\langle 0| \otimes I + \frac{1}{100} Z \otimes |1\rangle\langle 1| \otimes I, \\
h_{2,3} &= I \otimes |0\rangle\langle 0| \otimes X + \frac{1}{100} I \otimes |1\rangle\langle 1| \otimes Z, \\
h_{1,3} &= X \otimes I \otimes X.
\end{aligned}$$

Figure 3: A simple triangle Hamiltonian on three qubits.

the partitioning and snapping steps are trivial. The only nontrivial part of the procedure is the gapped rounding step, in which we choose a pivot operator A_i for each qubit i and then pinch the Hamiltonian terms with respect to these pivots.

We begin with qubit 2. The terms acting on qubit 2 are already written in the appropriate form, with the Pauli operators placed on the non-shared qubits 1 and 3:

$$\begin{aligned}
h_{1,2} &= (|0\rangle\langle 0|)_2 \otimes X_1 + \frac{1}{100} (|1\rangle\langle 1|)_2 \otimes Z_1, \\
h_{2,3} &= (|0\rangle\langle 0|)_2 \otimes X_3 + \frac{1}{100} (|1\rangle\langle 1|)_2 \otimes Z_3.
\end{aligned}$$

The coefficient $|0\rangle\langle 0|$ is gapped, so we take

$$A_2 = (|0\rangle\langle 0|)_2.$$

Next consider qubit 1. The term $h_{1,3}$ is already expressed in the desired form, while $h_{1,2}$ must be rewritten so that the Pauli operators act on the non-shared qubit 2:

$$\begin{aligned}
h_{1,2} &= \frac{1}{2} \left(X + \frac{1}{100} Z \right)_1 \otimes I_2 + \frac{1}{2} \left(X - \frac{1}{100} Z \right)_1 \otimes Z_2, \\
h_{1,3} &= X_1 \otimes X_3.
\end{aligned}$$

There are several possible gapped choices for the pivot. For simplicity, we take

$$A_1 = X_1.$$

Finally, for qubit 3, we proceed in the same way:

$$\begin{aligned}
h_{2,3} &= \frac{1}{2} I_2 \otimes \left(X + \frac{1}{100} Z \right)_3 + \frac{1}{2} Z_2 \otimes \left(X - \frac{1}{100} Z \right)_3, \\
h_{1,3} &= X_1 \otimes X_3.
\end{aligned}$$

Again, we choose the convenient pivot

$$A_3 = X_3.$$

With these choices of pivots, the pinching step is especially transparent: it forces the Hamiltonian to be diagonal in the tensor-product basis defined by the X basis on qubit 1, the standard basis on qubit 2, and the X basis on qubit 3. The rounded Hamiltonian

$$\hat{H} = \hat{h}_{1,2} + \hat{h}_{1,3} + \hat{h}_{2,3}$$

is therefore given by

$$\begin{aligned}
\hat{h}_{1,2} &= X_1 \otimes |0\rangle\langle 0|_2, \\
\hat{h}_{1,3} &= X_1 \otimes X_3, \\
\hat{h}_{2,3} &= |0\rangle\langle 0|_2 \otimes X_3.
\end{aligned}$$

These three terms now commute exactly.

The role of transitivity. This simple example also illustrates the role played by *transitivity* of commutation for 2×2 matrices. As discussed above, transitivity is what allows the rounding procedure to control commutation relations among different operators appearing in the Pauli decomposition of a single Hamiltonian term. In the present example, this control is reflected in the fact that the small Z -components in $h_{1,2}$ and $h_{2,3}$ are compatible with the choice of X -basis pivots on qubits 1 and 3.

One way to see this is to ask what would happen if the coefficient $1/100$ in $h_{1,2}$ were replaced by 1. Then $h_{1,2}$ and $h_{2,3}$ would still commute exactly, since they would remain diagonal in the same basis on the shared qubit 2. However, the commutator between $h_{1,2}$ and $h_{1,3}$ would become large. In fact, $h_{1,2}$ would no longer commute with any operator on qubits 1 and 3 that acts nontrivially on both qubits. Thus the smallness of the noncommuting component is essential: it is precisely what permits the global choice of pivots to round all three terms simultaneously.

Generalizing to m Hamiltonian terms. The same mechanism extends from this triangle example to general 2-local qubit Hamiltonians with m terms. Rather than choosing a rounding separately for each pair of Hamiltonian terms, the algorithm chooses pivots at the level of individual qubits. Each Hamiltonian term is then pinched with respect to the pivots on the qubits it touches. This one-shot choice is what preserves locality while enforcing exact commutation globally: at each qubit shared by two terms, their local components are diagonal in the same pivot basis, so every pair of rounded terms now commutes exactly.

The example above captures the main reason such a global procedure is needed. Even when two terms commute exactly, keeping them fixed may be incompatible with rounding the rest of the Hamiltonian. The rounding algorithm therefore treats the Hamiltonian as a whole, using the approximately commuting structure to identify local bases that are simultaneously compatible across all interactions.

1.5 Discussion and Future Work

The most immediate next question to investigate is whether the ε -dependence of our approximation result can be improved to match our lower bound of $\Omega(\sqrt{\varepsilon})$. Preliminary investigations using AI tools show that a refined application of our pinching and snapping techniques can indeed achieve an $O(\sqrt{\varepsilon})$ rounding error, but we leave the task of verifying this claim and generating a human-readable proof of it for future work.

After improving the ε dependence, the next natural questions to investigate are whether our rounding can be extended to systems of higher locality or local dimension. We suspect that going beyond qubits may be challenging, since the transitivity property that is crucial to our proof is no longer true for higher dimensions (even for dimension 3).

Another direction for generalization is to study looser notions of approximate commutation. For instance, what can we say if we only have a bound on the *average* norm of the commutators? Similar regimes were studied algorithmically in [SLS⁺25].

In the context of complexity theory, one interesting direction is to see what our result implies about the quantum PCP conjecture. Recall that our rounding result already implies that any qPCP-hard family of 2-local qubit Hamiltonians must exhibit some moderate amount of noncommutativity: assuming $\text{NP} \neq \text{QMA}$, any family of 2-local qubit Hamiltonians for which the γ -LH problem is QMA-hard for *constant* γ must allow for at least one of the pairwise commutator norms to be $\Omega(1)$. (In fact, such a family must also allow for a constant fraction of the terms to fail to approximately commute with the other terms—otherwise, these “bad terms” could simply be deleted from the Hamiltonian.) It

would be interesting to see whether this bound can be improved, for instance by finding a rounding technique that is more sensitive to the commutator norms of individual pairs of terms, rather than merely the maximum commutator norm.

Taking an alternate perspective, the complexity of the *general* k -local commuting Hamiltonian problem on qudits is wide open, and it is possible that some version of this problem is even qPCP-hard. Moreover, it is possible that gap-amplifying transformations for commuting Hamiltonians will be easier to design than for general Hamiltonians. Could some rounding scheme along the lines of ours show that gap amplification for commuting Hamiltonians directly implies the quantum PCP conjecture?

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The majority of this work was completed between September 2025 and February 2026 without substantial AI usage. We used ChatGPT for support with our literature review and the design of our figures (from September 2025 - February 2026), for pointing out a small modification to our proof to improve the error bound from $\varepsilon^{1/6}$ to $\varepsilon^{1/3}$ (in September 2026), for providing the example in section A (in September 2026), and for help with final editing (in September 2026); all other technical results and all the writing in this paper were human generated.

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2 Preliminaries

The d -dimensional complex vector space is denoted by $\mathbb{C}^d$. We use $L(V, W)$ to denote the set of linear operators mapping between finite-dimensional complex vector spaces V and W , and we use $L(\mathbb{C}^d)$ to denote the set of linear operators over $\mathbb{C}^d$. The *operator norm* of a linear operator $A \in L(V, W)$, for some finite-dimensional complex vector spaces $V \neq \{0\}$ and W is given by

$$\|A\| = \sup_{\|x\|_2=1} \|Ax\|_2, \quad (22)$$

where $\|x\|_2 = \sqrt{\langle x, x \rangle}$ is the standard Euclidean norm; we occasionally omit the subscript for convenience. The operator norm satisfies the triangle inequality and is submultiplicative with $\|A \cdot B\| \leq \|A\| \cdot \|B\|$, for all $A \in L(V, W)$ and $B \in L(U, V)$, where U is a finite-dimensional complex vector space. A linear operator $A \in L(\mathbb{C}^d)$ is called *positive semi-definite* (PSD), denoted by $A \succeq 0$, whenever $x^\dagger A x \geq 0$, for all $x \in \mathbb{C}^d$, and where $\dagger$ is the *adjoint*. A *unitary* $U : \mathbb{C}^d \rightarrow \mathbb{C}^d$ is a linear operator such that $U^\dagger U = U U^\dagger = I$, where I is the identity matrix on $\mathbb{C}^d$. A linear operator $H \in L(\mathbb{C}^d)$ is called *Hermitian* if $H^\dagger = H$. We use $\text{Herm}(\mathbb{C}^d)$ to denote the set of d -dimensional Hermitian matrices. A Hermitian operator $\Pi \in \text{Herm}(\mathbb{C}^d)$ is called a *projector* if $\Pi^2 = \Pi$. Any $A \in \text{Herm}(\mathbb{C}^2)$ admits a spectral decomposition of the form

$$A = \lambda_1 \Pi + \lambda_2 (I - \Pi) \quad (23)$$

where $\lambda_1, \lambda_2 \in \mathbb{R}$ are the eigenvalues of A , and where Π and $(I - \Pi)$ are the projectors onto the respective eigenspaces of A ; we call the eigenvalue difference

$$\Delta(A) := |\lambda_1 - \lambda_2| \quad (24)$$

the *spectral gap* of A , and we say that A is η -gapped if $\Delta(A) \geq \eta$.

Commutator. Let $A, B \in \mathbb{C}^{d \times d}$. Then, the *commutator* between A and B is defined as

$$[A, B] := AB - BA. \quad (25)$$

We say that A *commutes* with B (equivalently, B commutes with A) if $[A, B] = 0$ or $AB = BA$. For a specified $\varepsilon \geq 0$, we say that A and B ε -almost commute if

$$\|[A, B]\| \leq \varepsilon, \quad (26)$$

where $\|\cdot\|$ is the operator norm. We use the following elementary fact about the commutator.

Fact 2.1. Let $A, B, C \in \mathbb{C}^{d \times d}$ be arbitrary matrices. Then,

$$\|[A, C]\| \leq 2 \cdot \|A - B\| \cdot \|C\| + \|[B, C]\|. \quad (27)$$

Proof. First, notice that the commutator is linear in the sense that

$$[A, C] = [A - B, C] + [B, C]. \quad (28)$$

Using the triangle inequality and the sub-multiplicativity of the operator norm, we get

$$\|[A, C]\| = \|[A - B, C] + [B, C]\| \quad (29)$$

$$\leq \|[A - B, C]\| + \|[B, C]\| \quad (30)$$

$$= \|(A - B)C - C(A - B)\| + \|[B, C]\| \quad (31)$$

$$\leq \|(A - B)C\| + \|C(A - B)\| + \|[B, C]\| \quad (32)$$

$$\leq 2 \cdot \|A - B\| \cdot \|C\| + \|[B, C]\|. \quad (33)$$

□

We use the following well-known fact about *commuting* Hermitian matrices.

Fact 2.2 (Simultaneous Diagonalization). Let $A, B \in \text{Herm}(\mathbb{C}^2)$ be exactly commuting Hermitian matrices. Then, A and B can be simultaneously diagonalized in the following sense: there exists a spectral decomposition via common eigenspace projectors Π and $(I - \Pi)$ such that

$$A = a_1\Pi + a_2(I - \Pi) \quad \text{and} \quad B = b_1\Pi + b_2(I - \Pi) \quad (34)$$

where $a_1, a_2 \in \mathbb{R}$ and $b_1, b_2 \in \mathbb{R}$ are the eigenvalues of A and B , respectively.

Spectral stability. We use Weyl's inequality (e.g., see [Tao10]) which states that the eigenvalue spectra of Hermitian matrices remain stable under perturbations.

Lemma 2.3 (Weyl's inequality). Let $A, B \in \text{Herm}(\mathbb{C}^d)$. Then, for all $i \in [d]$,

$$|\lambda_i(A + B) - \lambda_i(A)| \leq \|B\|,$$

where $\lambda_i(A)$ and $\lambda_i(A + B)$ denote the i -th eigenvalues of A and $A + B$, respectively, listed in the same nondecreasing order.

The following is a direct consequence of Weyl's inequality in Lemma 2.3.

Fact 2.4. Let $A, B \in \text{Herm}(\mathbb{C}^d)$ and $\varepsilon \geq 0$. If $\|A - B\| \leq \varepsilon$, then $|\lambda_{\min}(A) - \lambda_{\min}(B)| \leq \varepsilon$.

Quantum information. A finite-dimensional complex Hilbert space is denoted by $\mathcal{H}$, and we use subscripts to distinguish between different systems (or registers); for example, we let $\mathcal{H}_1$ be the Hilbert space corresponding to the system labeled as 1. The tensor product of two Hilbert spaces $\mathcal{H}_1$ and $\mathcal{H}_2$ is another Hilbert space which we denote by $\mathcal{H}_{1,2} = \mathcal{H}_1 \otimes \mathcal{H}_2$.

A quantum system over the 2-dimensional Hilbert space $\mathcal{H} \cong \mathbb{C}^2$ is called a *qubit*; in general, when $\mathcal{H} \cong \mathbb{C}^d$ for $d \geq 2$, we call it a *qudit*. For $n \in \mathbb{N}$, we refer to quantum registers over the Hilbert space $\mathcal{H} \cong (\mathbb{C}^2)^{\otimes n}$ as *n-qubit states*. We use the word *quantum state* to refer to both pure states (unit vectors $|\psi\rangle \in \mathcal{H}$), as well as density matrices $\rho \in \mathcal{D}(\mathcal{H})$ with

$$\mathcal{D}(\mathcal{H}) = \{\rho \in \mathcal{L}(\mathcal{H}) : \rho \text{ is Hermitian, } \rho \geq 0 \text{ and } \text{Tr}[\rho] = 1\}. \quad (35)$$

Given a bipartite operator $\rho_{1,2} \in \mathcal{D}(\mathcal{H}_{1,2})$, we denote the partial trace over a system, say the second system, by $\text{Tr}_2[\cdot]$, and we call $\rho_1 = \text{Tr}_2[\rho_{1,2}]$ the reduced state. We use the following simple fact:

Fact 2.5. *Let $\mathcal{H}_{1,2} = \mathcal{H}_1 \otimes \mathcal{H}_2$ be a bipartite Hilbert space, for two finite-dimensional complex Hilbert spaces $\mathcal{H}_1$ and $\mathcal{H}_2$ of dimension d_1 and d_2 , respectively. Then, it holds that*

$$\|\rho_{1,2}\| \geq \frac{1}{d_2} \|\text{Tr}_2(\rho_{1,2})\|, \quad \forall \rho_{1,2} \in \mathcal{D}(\mathcal{H}_{1,2}). \quad (36)$$

Proof. By the variational characterization of the operator norm, we have

$$\begin{aligned} \|\rho_{1,2}\| &= \max_{\tau_{1,2} \in \mathcal{D}(\mathcal{H}_{1,2})} \text{Tr}(\rho_{1,2} \tau_{1,2}) \\ &\geq \max_{\sigma_1 \in \mathcal{D}(\mathcal{H}_1)} \text{Tr}\left(\rho_{1,2} \left(\sigma_1 \otimes \frac{I_2}{d_2}\right)\right) \\ &= \max_{\sigma_1 \in \mathcal{D}(\mathcal{H}_1)} \frac{1}{d_2} \text{Tr}(\text{Tr}_2(\rho_{1,2}) \sigma_1) \\ &= \frac{1}{d_2} \max_{\sigma_1 \in \mathcal{D}(\mathcal{H}_1)} \text{Tr}(\text{Tr}_2(\rho_{1,2}) \sigma_1) \\ &= \frac{1}{d_2} \|\text{Tr}_2(\rho_{1,2})\|. \end{aligned}$$

In the third line, we used the standard identity $\text{Tr}(\rho_{12}(\sigma_1 \otimes I_2)) = \text{Tr}(\text{Tr}_2(\rho_{12}) \sigma_1)$ to reduce the partial trace explicitly. This completes the proof. $\square$

Pauli matrices. The four 2×2 *Pauli matrices* are denoted as

$$\sigma^{(0)} = I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \sigma^{(1)} = X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (37)$$

$$\sigma^{(2)} = Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^{(3)} = Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (38)$$

The Pauli matrices are Hermitian and unitary, hence involutions: $(\sigma^{(\alpha)})^2 = I$. Each can therefore be viewed as an *observable*; for $\alpha = 1, 2, 3$ its eigenvalues are ± 1 .

Fact 2.6 (Properties of the Pauli matrices). *The Pauli matrices $\{\sigma^{(\alpha)}\}_{\alpha=0}^3$ satisfy*

$$\text{Tr}[\sigma^{(0)}] = 2, \quad \text{Tr}[\sigma^{(\alpha)}] = 0, \quad \forall \alpha = 1, 2, 3, \quad \text{and} \quad (39)$$

$$\text{Tr}[\sigma^{(\alpha)} \sigma^{(\beta)}] = 2\delta_{\alpha,\beta}, \quad \forall \alpha, \beta = 0, 1, 2, 3, \quad (40)$$

and form an orthogonal basis of $\mathbb{C}^{2 \times 2}$ with respect to the Hilbert-Schmidt inner product; in other words, any linear operator $A \in \mathbb{C}^{2 \times 2}$ can be written as a complex linear combination of the form

$$A = \frac{1}{2} \sum_{\alpha=0}^3 \text{Tr} [\sigma^{(\alpha)} A] \cdot \sigma^{(\alpha)}. \quad (41)$$

2.1 Local Hamiltonian Problem

We begin by recalling the promise-problem formulation of the local Hamiltonian problem. This language is convenient because local Hamiltonian problems naturally come with an energy gap promise: an instance is assumed to be either below a lower threshold or above an upper threshold, with no requirement on what happens in between.

Definition 2.7 (Binary Promise Problem). *Let $\{0,1\}^*$ be the set of all binary strings. $A = (A_{\text{YES}}, A_{\text{NO}})$ is called a binary promise problem when $A_{\text{YES}} \subseteq \{0,1\}^*$, $A_{\text{NO}} \subseteq \{0,1\}^*$, and $A_{\text{YES}} \cap A_{\text{NO}} = \emptyset$.*

We now specialize this general notion to the 2-local Hamiltonian problem. Throughout the paper, we keep track of both the absolute promise gap and the relative promise gap, since the latter is the natural normalization when the Hamiltonian has m local terms.

Definition 2.8 (2-Local Hamiltonian Problem γ -2-LH). *The 2-local Hamiltonian problem notated as γ -2-LH is a binary promise problem where the input is a classical binary string $x = (H, a, b)$ such that:*

- H is a 2-local Hamiltonian $H = \sum_{(i,j) \in E} h_{i,j}$ on a total of n qubits where $E \subseteq [n] \times [n]$ and $|E| = m = \text{poly}(n)$ and each $h_{i,j}$ is a Hermitian matrix with a bounded operator norm $\|h_{i,j}\| \leq 1$ and its entries are specified by $\text{poly}(n)$ bits and $h_{i,j}$ is non-identity on at most 2 qubits (namely i, j),
- a and b are rational or constant-degree real algebraic numbers with exact $\text{poly}(n)$ -bit descriptions such that $a < b$; the gap $\Gamma = b - a$ is called the **absolute promise gap** and $\gamma = \Gamma/m$ is called the **relative promise gap**, and we require $\gamma \geq 1/\text{poly}(n)$,
- for yes-instances, there exists an n -qubit quantum state $|\psi\rangle$ such that $\langle \psi | H | \psi \rangle \leq a$ (i.e. energy of the state w.r.t. H is at most a),
- for no-instances, for every n -qubit quantum state $|\psi\rangle$, it holds that $\langle \psi | H | \psi \rangle \geq b$ (i.e. energy of the state w.r.t. H is at least b), and
- it is promised that any instance will be either a yes or no instance.

The main objects of this paper are restricted versions of the local Hamiltonian problem in which the local terms commute, or nearly commute, with one another. The exactly commuting case serves as the classical benchmark against which we compare the almost commuting setting.

Definition 2.9 (Commuting 2-Local Hamiltonian Problem). *The Commuting Local Hamiltonian Problem notated as γ -2-CLH is defined as the γ -2-LH problem in Definition 2.8 with an additional restriction on the input that any pair of terms commute; in other words,*

$$[h_{i,j}, h_{k,\ell}] = 0, \quad \forall (i,j) \in E, \forall (k,\ell) \in E. \quad (42)$$

We also introduce the almost commuting analogue, where exact commutation is relaxed to a uniform operator-norm bound on all pairwise commutators. This is the promise problem to which our rounding theorem will later be applied.

Definition 2.10 (Almost Commuting Local Hamiltonian Problem). *The Almost Commuting 2-Local Hamiltonian Problem notated as (γ, ε) -2-ACLH is defined as the γ -2-LH problem in Definition 2.8 with an additional restriction that any pair of terms ε -almost-commute:*

$$\|[h_{i,j}, h_{k,\ell}]\| \leq \varepsilon, \quad \forall (i, j) \in E, \forall (k, \ell) \in E. \quad (43)$$

2.2 Complexity Classes

We next recall the complexity-theoretic notions used to state our containment and hardness consequences. Since the local Hamiltonian problem is naturally a promise problem, we use the corresponding promise-problem formulations of NP and QMA.

Definition 2.11 (NP). *A binary promise problem⁹ $A = (A_{\text{YES}}, A_{\text{NO}})$ is contained in NP if there exist polynomials $p(n), q(n)$ and a deterministic verifier V such that on instance $x \in A_{\text{YES}} \cup A_{\text{NO}}$ with $|x| = n$:*

1. **Completeness:** $x \in A_{\text{YES}} \implies \exists w : |w| = q(n)$ and $V(x, w)$ outputs 1 in $p(n)$ time; and
2. **Soundness:** $x \in A_{\text{NO}} \implies \forall w : V(x, w)$ outputs 0 in $p(n)$ time.

In contrast to NP, the class QMA allows the verifier to receive a quantum witness and to perform a quantum verification procedure. This is the natural quantum analogue of NP and is the complexity class captured by the general local Hamiltonian problem.

Definition 2.12 (QMA). *A promise problem $A = (A_{\text{YES}}, A_{\text{NO}})$ is contained in QMA if there exist polynomials $p(n), q(n), r(n)$ and a uniform family of quantum verifiers $\{V_n\}$ (that run in $p(n)$ time) such that on instance $x \in A_{\text{YES}} \cup A_{\text{NO}}$ with $|x| = n$:*

1. **Completeness:** If x is a yes-instance, i.e. $x \in A_{\text{YES}}$, then there exists a $q(n)$ -qubit quantum state $|\psi\rangle$ such that

$$\Pr[V_n(|x\rangle \otimes |0\rangle^{\otimes r(n)} \otimes |\psi\rangle) \text{ accepts}] \geq 2/3, \text{ and} \quad (44)$$

2. **Soundness:** If x is a no-instance, i.e. $x \in A_{\text{NO}}$, then for any $q(n)$ -qubit quantum state $|\psi\rangle$,

$$\Pr[V_n(|x\rangle \otimes |0\rangle^{\otimes r(n)} \otimes |\psi\rangle) \text{ accepts}] \leq 1/3. \quad (45)$$

We will also use the standard notion of completeness for promise-problem complexity classes. In our setting, completeness is used to formalize the sense in which the local Hamiltonian problem captures the full difficulty of QMA.

Definition 2.13 (Completeness). *We say that a promise problem $A = (A_{\text{YES}}, A_{\text{NO}})$ is complete for the class $\mathcal{C}$, notated as $\mathcal{C}$ -complete if:*

1. $A \in \mathcal{C}$; and

⁹We technically define a promise version of NP. Historically, containment in NP was defined when any binary string was either yes or no. That promise version of NP has been recently widely adopted when talking about promise problems.

2. for any promise problem $B = (B_{\text{YES}}, B_{\text{NO}}) \in \mathcal{C}$, there exists a polynomial-time reduction f such that:

- (a) $x \in B_{\text{YES}} \implies f(x) \in A_{\text{YES}}$; and
- (b) $x \in B_{\text{NO}} \implies f(x) \in A_{\text{NO}}$.

Finally, we record the standard hardness statement for the 2-local Hamiltonian problem in the relative-gap notation used throughout this paper. This result provides the baseline QMA-hardness result against which our NP containment for almost commuting Hamiltonians should be compared.

Theorem 2.14 ([KKR05]). *The γ -2-LH problem is QMA-complete when $\gamma = O(m^{-c})$ for any $c > 0$.*

Proof. The result shown in [KKR05] is that the problem is QMA-complete for some particular c , but a simple gap-amplification argument¹⁰ gives the claimed statement. We illustrate this argument with some concrete numbers. Suppose LH is QMA-hard for a relative gap of $\gamma = m^{-3}$, meaning an absolute gap in energy of $m\gamma$. Then consider a k -fold amplified Hamiltonian acting on a system of kn qubits divided into k groups of n qubits, given by $H_k = \sum_{i=1}^k (H)_i \otimes I_{\bar{i}}$. (That is, H_k is a sum of copies of H acting on the i th group of n qubits, for i running from 1 to k .) This Hamiltonian has $m' = km$ terms and an absolute energy gap of $km\gamma$, so the new relative gap is $\gamma'(m') = m^{-3} = (m'/k)^{-3}$. If we set $k = m^{99}$ —which gives $m' = m^{100}$ and $k = m'^{99/100}$ —then $m'/k = m'^{0.01}$, so $\gamma'(m') = (m')^{-0.03}$. This establishes that the LH problem for $\gamma = m^{-0.03}$ is QMA-hard. $\square$

Theorem 2.15 ([BV04]). *The γ -2-CLH promise problem is contained in NP for any admissible¹¹ γ .*

Proof. Lemma 5 of [BV04] shows that the problem γ -2-CLH for any value of γ is contained in NP if the "common eigenspace problem" (CES) for 2-local projectors is contained in NP. This latter containment is shown in Theorem 3 of [BV04]. $\square$

3 Commutativity in the World of Qubit Operators

In this section, we collect the qubit-specific facts that make the pinching step possible. We first define the pinching and snapping operations for single-qubit operators, and then prove the elementary norm bounds that control the errors they introduce. We also isolate the transitivity phenomenon for commuting qubit operators, which is one of the main algebraic features that fails in higher local dimension and is used crucially in our rounding argument.

3.1 Pinching by a Hermitian Qubit Operator

Definition 3.1 (Pinching operator). *Let $A \in \text{Herm}(\mathbb{C}^2)$ be Hermitian with $\Delta(A) > 0$ and spectral decomposition*

$$A = \lambda_{\min}(A) \Pi + \lambda_{\max}(A) (I - \Pi), \quad (46)$$

¹⁰We have heard it credited to Lijie Chen, but it may be folklore.

¹¹Our definition of γ -2-CLH requires $\gamma \geq 1/\text{poly}(n)$.

where Π and $(I - \Pi)$ are the eigenspace projectors. Then, the pinching operator $\mathcal{P}_A : \mathcal{L}(\mathbb{C}^2) \rightarrow \mathcal{L}(\mathbb{C}^2)$ associated with the eigenspace projectors of A is defined as

$$\mathcal{P}_A(X) = \Pi X \Pi + (I - \Pi)X(I - \Pi), \quad \text{for } X \in \mathbb{C}^{2 \times 2}. \quad (47)$$

For a scalar operator $A = cI$, we define $\mathcal{P}_A = \text{id}$.

Equipped with Definition 3.1, we now show the following result.

Lemma 3.2 (Pinching by a Hermitian qubit operator). *Let $A \in \text{Herm}(\mathbb{C}^2)$ be a Hermitian operator with $\Delta(A) > 0$ and spectral decomposition given by*

$$A = \lambda_{\min}(A) \Pi + \lambda_{\max}(A) (I - \Pi) \quad (48)$$

with eigenvalues $\lambda_{\min}, \lambda_{\max} \in \mathbb{R}$ and eigenspace projectors Π and $(I - \Pi)$. Let $\mathcal{P}_A$ be the pinching operator in Definition 3.1 with respect to A . Then, for any $B \in \mathbb{C}^{2 \times 2}$, the pinched matrix $\mathcal{P}_A(B)$ satisfies

$$[\mathcal{P}_A(B), A] = 0 \quad \text{and} \quad \|B - \mathcal{P}_A(B)\| = \frac{\|[B, A]\|}{\Delta(A)}, \quad (49)$$

where $\Delta(A) = \lambda_{\max}(A) - \lambda_{\min}(A)$ is the spectral gap of A .

Proof. Both the operator norm and the commutator are invariant under unitary conjugation. Hence, without loss of generality, we may assume that A is diagonal in its eigenbasis:

$$A = \begin{pmatrix} \lambda_{\min}(A) & 0 \\ 0 & \lambda_{\max}(A) \end{pmatrix} \quad \text{and} \quad \Pi = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}. \quad (50)$$

Expanding B in the eigenbasis of A , we get

$$B = \begin{pmatrix} B_{1,1} & B_{1,2} \\ B_{2,1} & B_{2,2} \end{pmatrix}. \quad (51)$$

A direct computation shows that

$$[B, A] = BA - AB \quad (52)$$

$$= (\lambda_{\max}(A) - \lambda_{\min}(A)) \begin{pmatrix} 0 & B_{1,2} \\ -B_{2,1} & 0 \end{pmatrix} = \Delta(A) \cdot \begin{pmatrix} 0 & B_{1,2} \\ -B_{2,1} & 0 \end{pmatrix}. \quad (53)$$

By definition of the pinching operator, we have

$$\mathcal{P}_A(B) = \begin{pmatrix} B_{1,1} & 0 \\ 0 & B_{2,2} \end{pmatrix}, \quad (54)$$

which is diagonal in the eigenbasis of A , and hence $[\mathcal{P}_A(B), A] = 0$. Moreover,

$$B - \mathcal{P}_A(B) = \begin{pmatrix} 0 & B_{1,2} \\ B_{2,1} & 0 \end{pmatrix}. \quad (55)$$

Next, consider the Pauli- Z unitary

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (56)$$

It is now easy to see that

$$B - \mathcal{P}_A(B) = \Delta(A)^{-1} \cdot Z \cdot [B, A]. \quad (57)$$

Using the unitary invariance of the operator norm with respect to Z , we obtain the equality

$$\|B - \mathcal{P}_A(B)\| = \frac{\|Z \cdot [B, A]\|}{\Delta(A)} = \frac{\|[B, A]\|}{\Delta(A)}. \quad (58)$$

□

3.2 Snapping Hermitian qubit operators to multiples of the Identity

In this section, we formally introduce the mathematical background for the “snapping technique”.

Definition 3.3 (Snapping operation). *The snapping operation is the linear map*

$$\mathcal{S} : \text{Herm}(\mathbb{C}^2) \rightarrow \text{Herm}(\mathbb{C}^2), \quad X \mapsto \mathcal{S}(X) := \frac{\text{Tr}(X)}{2} I. \quad (59)$$

For a Hermitian qubit operator, $\text{Tr}(X)/2 = (\lambda_{\min}(X) + \lambda_{\max}(X))/2$ is the midpoint of its spectrum.

Lemma 3.4 (Snapping a Hermitian qubit operator to the identity). *Let $B \in \text{Herm}(\mathbb{C}^2)$ be any matrix with spectral decomposition*

$$B = \lambda_{\min}(B) \Pi + \lambda_{\max}(B) (I - \Pi), \quad (60)$$

and let $\mathcal{S}$ denote the snapping operator defined in Definition 3.3. Then,

$$\|B - \mathcal{S}(B)\| = \frac{\lambda_{\max}(B) - \lambda_{\min}(B)}{2} = \frac{\Delta(B)}{2}, \quad \|\mathcal{S}(B)\| \leq \|B\|, \quad (61)$$

where $\Delta(B) = \lambda_{\max}(B) - \lambda_{\min}(B)$ is the spectral gap of B .

Proof. The operator norm is invariant under unitary conjugation. Hence, without loss of generality, we may assume that B is diagonal in its eigenbasis:

$$B = \begin{pmatrix} \lambda_{\min}(B) & 0 \\ 0 & \lambda_{\max}(B) \end{pmatrix}. \quad (62)$$

By definition of the snapping operation, with $c = (\lambda_{\min}(B) + \lambda_{\max}(B))/2$, we have

$$\mathcal{S}(B) = cI, \quad \mathcal{S}(B) - B = \begin{pmatrix} \Delta(B)/2 & 0 \\ 0 & -\Delta(B)/2 \end{pmatrix}. \quad (63)$$

The operator norm of a Hermitian matrix equals the maximum absolute value of its eigenvalues. Consequently,

$$\|B - \mathcal{S}(B)\| = \frac{\Delta(B)}{2}, \quad \|\mathcal{S}(B)\| = |c| \leq \max\{|\lambda_{\min}(B)|, |\lambda_{\max}(B)|\} = \|B\|. \quad (64)$$

□

3.3 Transitivity of (Almost) Commutativity of Qubit Operators

Lemma 3.5 (Gapped spectra imply transitivity of commutativity). *Let $A, B, C \in \text{Herm}(\mathbb{C}^2)$ and suppose that B has non-zero spectral gap $\Delta(B) = \lambda_{\max}(B) - \lambda_{\min}(B) > 0$. Then,*

$$[A, B] = 0 \quad \text{and} \quad [B, C] = 0 \quad \text{implies} \quad [A, C] = 0.$$

Proof. Since $\Delta(B) > 0$, the matrix B admits an eigenbasis, unique up to phases and ordering, in which it is diagonal. Let $\{\Pi, \Pi^\perp\}$ be the orthogonal projectors onto the respective eigenspaces. Because A commutes with B , and C commutes with B , we can

use simultaneous diagonalization (see Fact 2.2) and the fact¹² $\Delta(B) > 0$ to diagonalize all three matrices with respect to the same eigenspace projectors $\{\Pi, \Pi^\perp\}$ as follows:

$$A = a_1\Pi + a_2\Pi^\perp \quad (65)$$

$$B = b_1\Pi + b_2\Pi^\perp \quad (66)$$

$$C = c_1\Pi + c_2\Pi^\perp \quad (67)$$

Using the orthogonality of $\{\Pi, \Pi^\perp\}$, we then find that

$$\begin{aligned} [A, C] &= [a_1\Pi + a_2\Pi^\perp, c_1\Pi + c_2\Pi^\perp] \\ &= a_1c_1[\Pi, \Pi] + a_1c_2[\Pi, \Pi^\perp] + a_2c_1[\Pi^\perp, \Pi] + a_2c_2[\Pi^\perp, \Pi^\perp] = 0. \end{aligned}$$

□

The exact transitivity statement above is useful only after the relevant operators have been made exactly commuting. In the rounding argument, however, we typically know only that two local components almost commute with a common gapped pivot. The following lemma provides the corresponding robust version: if A and C each almost commute with a sufficiently gapped qubit operator B , then A and C must almost commute with each other.

Lemma 3.6 (Gapped spectra imply transitivity of almost-commutativity). *Let $A, B, C \in \text{Herm}(\mathbb{C}^2)$ and suppose that B has non-zero gap $\Delta(B) = \lambda_{\max}(B) - \lambda_{\min}(B) \geq \xi > 0$. Then,*

$$\|[A, B]\| \leq \varepsilon \quad \text{and} \quad \|[B, C]\| \leq \varepsilon \quad \text{implies} \quad \|[A, C]\| \leq \frac{2\varepsilon}{\xi} \left(\|A\| + \|C\| + \frac{\varepsilon}{\xi} \right).$$

When $\|A\| \leq 1$, $\|C\| \leq 1$, and $\xi \geq \varepsilon > 0$, the bound becomes

$$\|[A, C]\| \leq \frac{4\varepsilon}{\xi} + \frac{2\varepsilon^2}{\xi^2} \leq \frac{6\varepsilon}{\xi}.$$

Proof. According to Lemma 3.2, there exist (via pinching by B) A' and C' such that $[A', B] = [B, C'] = 0$ and $\|A - A'\| \leq \frac{\varepsilon}{\xi}$ and $\|C - C'\| \leq \frac{\varepsilon}{\xi}$. By Lemma 3.5, it holds that $[C', A'] = 0$. Thus, we can bound $\|[A, C]\|$ as follows:

$$\begin{aligned} \|[A, C]\| &\leq 2 \cdot \|A - A'\| \cdot \|C\| + \|[A', C]\| && \text{(via Fact 2.1)} \\ &\leq \frac{2\varepsilon}{\xi} \cdot \|C\| + \|[C, A']\| && (\|A - A'\| \leq \frac{\varepsilon}{\xi} \text{ and } [C, A'] = -[A', C]) \\ &\leq \frac{2\varepsilon}{\xi} \cdot \|C\| + 2 \cdot \|C - C'\| \cdot \|A'\| + \|[C', A']\| && \text{(via Fact 2.1)} \\ &\leq \frac{2\varepsilon}{\xi} \cdot \|C\| + \frac{2\varepsilon}{\xi} \cdot \|A'\| && (\|C - C'\| \leq \frac{\varepsilon}{\xi} \text{ and } [C', A'] = 0) \\ &\leq \frac{2\varepsilon}{\xi} (\|C\| + \|A\| + \varepsilon/\xi) && (\|A'\| \leq \|A\| + \varepsilon/\xi). \end{aligned}$$

□

¹²Without that condition, we can only conclude that A and B can be simultaneously diagonalized and that B and C can be simultaneously diagonalized, but not necessarily all three can be simultaneously diagonalized. If B is a multiple of the identity, then any two non-commuting matrices will be a counterexample.

4 Local-to-Global Framework for Operators on 2 Qubits

In this section, we develop the local-to-global tools used in the rounding argument. The main point is that approximate commutation of two-qubit Hamiltonian terms can be detected at the level of the single-qubit operators appearing in their local Pauli decompositions. This allows us to translate global commutator bounds into local algebraic constraints, which will later be used to choose pivots, perform snapping, and preserve locality throughout the rounding procedure.

4.1 Local Pauli Decomposition

The following simple fact gives a decomposition of a two-qubit operator acting on qubits s and r into a sum of tensor product operators. Similar decompositions play a deep role in the theory of commuting Hamiltonians, for instance in Equation (1) of [AE11].

Lemma 4.1 (Local Pauli decomposition). *Let $h_{s,r} \in L(\mathcal{H}_s \otimes \mathcal{H}_r)$ be a linear two-qubit operator acting on the qubits labeled by s and r . Then, $h_{s,r}$ can be decomposed as a linear combination of the form*

$$h_{s,r} = \sum_{\alpha=0}^3 A_s^{(\alpha)} \otimes \sigma_r^{(\alpha)} \quad (68)$$

where $\{\sigma_r^{(\alpha)}\}_{\alpha=0}^3$ are the Pauli matrices on the r -th qubit, and where $\{A_s^{(\alpha)}\}_{\alpha=0}^3$ is an ensemble of 2×2 complex matrices. Furthermore, the components $\{A_s^{(\alpha)}\}_{\alpha=0}^3$ are Hermitian if and only if $h_{s,r}$ is Hermitian.

Proof. Here, we make use of Fact 2.6; namely, that the Pauli matrices form an orthogonal basis of $L(\mathbb{C}^2)$ with respect to the Hilbert-Schmidt inner product. Observing that tensor products of Pauli matrices form a basis of the space of linear operators on the tensor system $\mathbb{C}^2 \otimes \mathbb{C}^2$, we obtain the decomposition

$$h_{s,r} = \sum_{\alpha=0}^3 \sum_{\beta=0}^3 c_{\alpha,\beta} \sigma_s^{(\beta)} \otimes \sigma_r^{(\alpha)}.$$

Next, we let $A_s^{(\alpha)} = \sum_{\beta=0}^3 c_{\alpha,\beta} \sigma_s^{(\beta)}$ and show the *if and only if* statement. First, if $A_s^{(\alpha)}$ is Hermitian, then its tensor product with a Hermitian Pauli matrix must also be Hermitian, and the sum $h_{s,r}$ over these tensor products is going to be Hermitian. Second, to show that if $h_{s,r}$ is Hermitian, then each $A_s^{(\alpha)}$ is Hermitian, we proceed as follows; first, observe that

$$\mathrm{Tr}_r \left[\left(I \otimes \sigma_r^{(\alpha)} \right) h_{s,r} \right] = \mathrm{Tr}_r \left[\sum_{\beta=0}^3 A_s^{(\beta)} \otimes \sigma_r^{(\alpha)} \sigma_r^{(\beta)} \right] \quad (\text{by Equation (68)}) \quad (69)$$

$$= \sum_{\beta=0}^3 A_s^{(\beta)} \cdot \mathrm{Tr} \left(\sigma_r^{(\alpha)} \sigma_r^{(\beta)} \right) \quad (\text{by linearity of the trace}) \quad (70)$$

$$= \sum_{\beta=0}^3 2\delta_{\alpha,\beta} A_s^{(\beta)} \quad (\text{by Fact 2.6}) \quad (71)$$

$$= 2A_s^{(\alpha)}. \quad (72)$$

We can similarly compute $\mathrm{Tr}_r \left[h_{s,r} \left(I \otimes \sigma_r^{(\alpha)} \right) \right] = 2A_s^{(\alpha)}$. Combining the two, we get

$$A_s^{(\alpha)} = \frac{1}{2} \mathrm{Tr}_r \left[\left(I \otimes \sigma_r^{(\alpha)} \right) h_{s,r} \right] = \frac{1}{2} \mathrm{Tr}_r \left[h_{s,r} \left(I \otimes \sigma_r^{(\alpha)} \right) \right]. \quad (73)$$

Putting everything together, we can verify that $A_s^{(\alpha)}$ is Hermitian by noting the following.

$$\left(A_s^{(\alpha)}\right)^\dagger = \frac{1}{2} \text{Tr}_r \left[\left(I \otimes \sigma_r^{(\alpha)} \right) h_{s,r} \right]^\dagger \quad (\text{by Equation (73)}) \quad (74)$$

$$= \frac{1}{2} \text{Tr}_r \left[h_{s,r}^\dagger \left(I \otimes \sigma_r^{(\alpha)} \right)^\dagger \right] \quad (\text{"Socks and Shoes" property}) \quad (75)$$

$$= \frac{1}{2} \text{Tr}_r \left[h_{s,r} \left(I \otimes \sigma_r^{(\alpha)} \right) \right] \quad (h_{s,r} \text{ and } \sigma_r^{(\alpha)} \text{ are hermitian}) \quad (76)$$

$$= A_s^{(\alpha)} \quad (\text{by Equation (73)}) \quad (77)$$

□

When working with local Pauli decompositions of two-qubit operators, it will be convenient for us to use the following notion of a *gapped* decomposition.

Definition 4.2 (Gapped Pauli decomposition). *A two-qubit operator $h_{s,r} \in \text{Herm}(\mathcal{H}_s \otimes \mathcal{H}_r)$ acting on the qubits labeled by s and r is said to be η -gapped on the s -subsystem (notated as (s, η) -gapped) if any of the local components $\{A_s^{(\alpha)}\}_{\alpha=0}^3$ which appear in the local Pauli decomposition of the form*

$$h_{s,r} = \sum_{\alpha=0}^3 A_s^{(\alpha)} \otimes \sigma_r^{(\alpha)} \quad (78)$$

is η -gapped; in other words, there exists a matrix $A_s^{(\alpha^)} \in \mathbb{C}^{2 \times 2}$ which is η -gapped such that $\Delta(A_s^{(\alpha^*)}) \geq \eta$. The property (r, η) -gapped is defined similarly. We say the operator is η -gapped on both subsystems if it is both (r, η) -gapped and (s, η) -gapped.*

4.2 Global-to-Local Inheritance of Norms of Commutators

Let $h_{s,r}$ and $h_{s,q}$ be 2 Hamiltonian terms on qubit pairs $\{s, r\}$ and $\{s, q\}$, where s, r, q are distinct, such that they ε -almost commute i.e. $\|[h_{s,r}, h_{s,q}]\| \leq \varepsilon$. Let their local components on the shared qubit be $\{A^{(\alpha)}\}$ and $\{B^{(\beta)}\}$. An important question for our techniques is: what can we say about the commutator of any pair $\|[A^{(\alpha)}, B^{(\beta)}]\|$? We will prove in Theorem 4.4 that any pair of local components must also ε -almost commute. For that proof, we will use Lemma 4.3.

Lemma 4.3 (Operator norm lower bound from orthogonal components). *Let $\{M_i\}_i \subseteq \mathbb{C}^{d_1 \times d_1}$, and let $\{N_i\}_i \subseteq \mathbb{C}^{d_2 \times d_2}$ be Hermitian binary observables satisfying*

$$N_i^2 = I_{d_2} \quad \text{and} \quad \text{Tr}[N_i N_j] = d_2 \delta_{i,j}. \quad (79)$$

Then, any operator of the form $T = \sum_i M_i \otimes N_i$ satisfies

$$\|T\| \geq \|M_i\|, \quad \text{for every } i. \quad (80)$$

Proof. We compute

$$T^\dagger T = \sum_{i,j} M_i^\dagger M_j \otimes N_i N_j, \quad (81)$$

where we used that each N_i is Hermitian. Taking the partial trace over the second subsystem and using the Hilbert-Schmidt orthogonality of the N_i 's gives

$$\text{Tr}_2[T^\dagger T] = \sum_{i,j} M_i^\dagger M_j \text{Tr}[N_i N_j] \quad (82)$$

$$= d_2 \sum_i M_i^\dagger M_i. \quad (83)$$

By Fact 2.5, extended by homogeneity to positive semidefinite operators, we have

$$\|T\|^2 = \|T^\dagger T\| \geq \frac{1}{d_2} \|\text{Tr}_2[T^\dagger T]\| = \left\| \sum_i M_i^\dagger M_i \right\|. \quad (84)$$

Since each $M_j^\dagger M_j$ is positive semidefinite, we have

$$\sum_j M_j^\dagger M_j \succeq M_i^\dagger M_i \quad (85)$$

for every i , and hence

$$\left\| \sum_j M_j^\dagger M_j \right\| \geq \|M_i^\dagger M_i\| = \|M_i\|^2. \quad (86)$$

Combining the two inequalities yields $\|T\|^2 \geq \|M_i\|^2$, and therefore $\|T\| \geq \|M_i\|$. $\square$

Theorem 4.4. *Let $h_{s,r}$ and $h_{s,q}$ be two 2-qubit linear operators on qubit pairs $\{s, r\}$ and $\{s, q\}$, where s, r, q are distinct. According to Lemma 4.1, these can be written as*

$$h_{s,r} = \sum_{\alpha=0}^3 A_s^{(\alpha)} \otimes \sigma_r^{(\alpha)} \quad h_{s,q} = \sum_{\beta=0}^3 B_s^{(\beta)} \otimes \sigma_q^{(\beta)} \quad (87)$$

for ensembles of 2×2 complex matrices $\{A_s^{(\alpha)}\}_{\alpha=0}^3$ and $\{B_s^{(\beta)}\}_{\beta=0}^3$. Then, if the terms satisfy $\|[h_{s,r}, h_{s,q}]\| \leq \varepsilon$, for some $\varepsilon \geq 0$, this implies that $\|[A_s^{(\alpha)}, B_s^{(\beta)}]\| \leq \varepsilon$, for all α, β .

Proof. We can re-write the linear operators with an identity factor on the unaffected qubit,

$$h_{s,r} = \sum_{\alpha=0}^3 A_s^{(\alpha)} \otimes \sigma_r^{(\alpha)} \otimes I_q \quad \text{and} \quad h_{s,q} = \sum_{\beta=0}^3 B_s^{(\beta)} \otimes I_r \otimes \sigma_q^{(\beta)}. \quad (88)$$

Then, we can compute the products $h_{s,r}h_{s,q}$ and $h_{s,q}h_{s,r}$ as follows.

$$h_{s,r}h_{s,q} = \sum_{\alpha, \beta} A_s^{(\alpha)} B_s^{(\beta)} \otimes \sigma_r^{(\alpha)} \otimes \sigma_q^{(\beta)} \quad (89)$$

$$h_{s,q}h_{s,r} = \sum_{\alpha, \beta} B_s^{(\beta)} A_s^{(\alpha)} \otimes \sigma_r^{(\alpha)} \otimes \sigma_q^{(\beta)} \quad (90)$$

Then, we can plug these in the definition of the commutator.

$$\begin{aligned} [h_{s,r}, h_{s,q}] &= h_{s,r}h_{s,q} - h_{s,q}h_{s,r} \\ &= \sum_{\alpha, \beta} \left(A_s^{(\alpha)} B_s^{(\beta)} - B_s^{(\beta)} A_s^{(\alpha)} \right) \otimes \sigma_r^{(\alpha)} \otimes \sigma_q^{(\beta)} \\ &= \sum_{\alpha, \beta} \left[A_s^{(\alpha)}, B_s^{(\beta)} \right] \otimes \left[\sigma_r^{(\alpha)} \otimes \sigma_q^{(\beta)} \right] \end{aligned}$$

Using Fact 2.6, we can conclude that $\{\sigma_r^{(\alpha)} \otimes \sigma_q^{(\beta)}\}_{\alpha, \beta}$ forms a set of pairwise orthogonal observables on the tensor product system $\mathcal{H}_r \otimes \mathcal{H}_q$. Thus, for each α, β , Lemma 4.3 implies that

$$\|[A_s^{(\alpha)}, B_s^{(\beta)}]\| \leq \|[h_{s,r}, h_{s,q}]\| \leq \varepsilon. \quad (91)$$

This proves the claim. $\square$

4.3 Snapping a 2-local Hamiltonian term into 1-local Hamiltonian term

Theorem 4.5 (Snapping 2-local Hamiltonian terms into 1-local terms). *Let $h_{s,r} \in \text{Herm}(\mathcal{H}_s \otimes \mathcal{H}_r)$ be a two-qubit operator acting on qubits s and r which is not (s, η) -gapped according to Definition 4.2. Then, there exists a 1-local Hamiltonian term $\hat{h}_r$ such that:*

$$\|h_{s,r} - I_s \otimes \hat{h}_r\| \leq 4\eta, \quad \|\hat{h}_r\| \leq \|h_{s,r}\|. \quad (92)$$

Proof. Using Lemma 4.1, we can expand the two-qubit operator $h_{s,r}$ as

$$h_{s,r} = \sum_{\alpha=0}^3 A_s^\alpha \otimes \sigma_r^{(\alpha)} = \sum_{\alpha=0}^3 \left(\lambda_{\max}^\alpha \Pi_s^\alpha + \lambda_{\min}^\alpha (I - \Pi_s^\alpha) \right) \otimes \sigma_r^{(\alpha)}.$$

The hypothesis that $h_{s,r}$ is not (s, η) -gapped means that for all α , $\Delta(A_s^\alpha) < \eta$. Applying the snapping operation of Definition 3.3, we replace each coefficient by the midpoint of its spectrum:

$$\hat{A}_s^\alpha := \mathcal{S}(A_s^\alpha) = \frac{\text{Tr}(A_s^\alpha)}{2} I_s = \frac{\lambda_{\min}(A_s^\alpha) + \lambda_{\max}(A_s^\alpha)}{2} I_s.$$

The induced map on the full two-qubit term is norm-nonincreasing. Define the operator on qubit r by

$$\hat{h}_r := \sum_{\alpha=0}^3 \frac{\text{Tr}(A_s^\alpha)}{2} \sigma_r^{(\alpha)} = \frac{1}{2} \text{Tr}_s(h_{s,r}). \quad (93)$$

Indeed, conjugating a single-qubit operator by each of I, X, Y, Z and averaging removes its traceless Pauli components. Thus,

$$I_s \otimes \hat{h}_r = \frac{1}{4} \sum_{\mu=0}^3 (\sigma_s^{(\mu)} \otimes I_r) h_{s,r} (\sigma_s^{(\mu)} \otimes I_r).$$

Each $\sigma_s^{(\mu)} \otimes I_r$ is unitary. By the triangle inequality and unitary invariance,

$$\|\hat{h}_r\| = \|I_s \otimes \hat{h}_r\| \leq \frac{1}{4} \sum_{\mu=0}^3 \|h_{s,r}\| = \|h_{s,r}\|.$$

Since $\|A_s^\alpha - \hat{A}_s^\alpha\| = \Delta(A_s^\alpha)/2$, we can calculate the rounding error explicitly as follows:

$$\begin{aligned} \|h_{s,r} - I_s \otimes \hat{h}_r\| &= \left\| \sum_{\alpha=0}^3 (A_s^\alpha - \hat{A}_s^\alpha) \otimes \sigma_r^{(\alpha)} \right\| \\ &\leq \sum_{\alpha=0}^3 \left\| (A_s^\alpha - \hat{A}_s^\alpha) \otimes \sigma_r^{(\alpha)} \right\| && \text{(triangle inequality)} \\ &\leq \sum_{\alpha=0}^3 \left\| A_s^\alpha - \hat{A}_s^\alpha \right\| \cdot \underbrace{\|\sigma_r^{(\alpha)}\|}_{=1} && \text{(by the properties of } \|\cdot\| \text{)} \\ &\leq \frac{1}{2} \sum_{\alpha=0}^3 \Delta(A_s^\alpha) \leq 2\eta \leq 4\eta. \end{aligned}$$

□

5 Rounding Almost Commuting 2-Local Qubit Hamiltonians

In this section, we prove the main result of our paper: any almost commuting 2-local Hamiltonian (on qubits) can be mapped to a nearby exactly commuting (qubit) Hamiltonian of the same locality. The proof of our main result uses a technical helper lemma (Lemma 5.2), which we formally state and prove later in this section.

Theorem 5.1 (Algorithmic rounding for almost commuting 2-local Hamiltonians on qubits).

Let $H = \sum_{I \in \mathcal{I}} h_I$ be a 2-local Hamiltonian on n qubits, where $\mathcal{I}$ is a collection of distinct sets of indices affected by each Hamiltonian term, where $m = |\mathcal{I}|$ is the total number of terms, and where

- each term h_I , for $I \in \mathcal{I}$, is a Hermitian operator that acts as non-identity on at most two qubits (i.e. $|I| \leq 2$), of unit norm $\|h_I\| \leq 1$, and
- all terms pair-wise ε -almost commute with $0 \leq \varepsilon \leq 1$ such that

$$\|[h_I, h_J]\| \leq \varepsilon, \quad \forall I, J \in \mathcal{I}.$$

Given a description of H and the bound ε , it is possible to compute in classical deterministic time $O(m)$ ¹³ the description of a nearby exactly commuting Hamiltonian $\hat{H} = \sum_{I \in \mathcal{I}} \hat{h}_I$ such that

- $\hat{H}$ is 2-local, thereby preserving the locality of the Hamiltonian H , and $\|\hat{h}_I\| \leq \|h_I\| \leq 1$ for every $I \in \mathcal{I}$,
- $\hat{H}$ is a qubit Hamiltonian, thereby preserving the local dimension of H ,
- all of the terms in $\hat{H}$ pairwise commute such that

$$[\hat{h}_I, \hat{h}_J] = 0, \quad \forall I, J \in \mathcal{I}, \tag{94}$$

- all of the terms in $\hat{H}$ are (point-wise) nearby in the sense that

$$\|\hat{h}_I - h_I\| \leq 200 \varepsilon^{1/3}, \quad \forall I \in \mathcal{I}, \text{ and} \tag{95}$$

- $\hat{H}$ is overall close to H in operator norm such that

$$\|\hat{H} - H\| \leq 200 m \varepsilon^{1/3}. \tag{96}$$

Proof. If $\varepsilon = 0$, set $\hat{H} = H$, which satisfies all the claims. Henceforth assume $\varepsilon > 0$. We remark that the statement of the theorem allows the Hamiltonian to have terms that have locality strictly less than 2. For convenience, we assume that the Hamiltonian is of the form $H = \sum_{(i,j) \in E} h_{i,j}$ so that all terms in H are formally two-local. This avoids additional bookkeeping and is purely for clarity of the exposition (it does not materially affect any of the arguments). The ordering of the two indices on the Hamiltonian term is irrelevant, so we will freely write $h_{i,j}$ or $h_{j,i}$ for the same term as convenient.

Our rounding procedure generates the Hamiltonian $\hat{H}$ as follows. We will have two stages of snapping with gap parameters η_2 in the first stage and η_1 in the second stage (note that the index is counting down), with $\eta_1 = \eta_2 = \varepsilon^{1/3}$. Similarly, the commutator norm will be bounded by ε_2 in the first stage and ε_1 in the second stage (with the index counting down as well).

¹³And polynomial in the bit complexity of each Hamiltonian term and of ε .

1. Define sets of terms T_2, T_1, T_0 that will keep track of 2-local, 1-local, and 0-local terms, respectively, through the snapping process. To start out, let T_2 contain all the terms in H , and let T_1 and T_0 be empty.

Let $\varepsilon_2 = \varepsilon$.

2. **Snapping 2-local terms.** Let $\eta_2 := \varepsilon_2^{1/3}$. For every term $h_{i,j}$ in T_2 :

- (a) If $h_{i,j}$ is η_2 -gapped on both subsystems according to Definition 4.2 (i.e. (i, η_2) -gapped and (j, η_2) -gapped), remove it from T_2 and place it in $H_2^{(\text{snap})}$.
- (b) Else, pick the qubit j (WLOG) such that $h_{i,j}$ is not (j, η_2) -gapped. Snap using Theorem 4.5 into a 1-local term on the subsystem i to obtain $h'_{i,j}$ as below and place $h'_{i,j}$ in T_1 .

$$h'_{i,j} = g_i \otimes I_j \quad \text{where} \quad \|h'_{i,j} - h_{i,j}\| \leq 4\eta_2. \quad (97)$$

3. **Snapping 1-local terms.** Let $\eta_1 := \eta_2$. For each term $h'_{i,j}$ in T_1 (1-local terms), perform the following.

- (a) If $h'_{i,j}$ is (i, η_1) -gapped, place the term into $H_1^{(\text{snap})}$.
- (b) Else, snap that term into a multiple of the identity $h''_{i,j} := \mathcal{S}(g_i) \otimes I_j = cI$, where $c = \text{Tr}(g_i)/2$, and place that term into T_0 .

$$\|h''_{i,j} - h'_{i,j}\| \leq 4\eta_1. \quad (98)$$

4. Move every term in T_0 to $H_0^{(\text{snap})}$.

5. We now have:

$$H^{(\text{snap})} = H_2^{(\text{snap})} + H_1^{(\text{snap})} + H_0^{(\text{snap})} \quad (99)$$

Both snapping stages are norm-nonincreasing: this follows from Theorem 4.5 for the first stage and from $\|\mathcal{S}(g_i)\| = |\text{Tr}(g_i)|/2 \leq \|g_i\|$ for the second. Thus $\|h_{i,j}^{(\text{snap})}\| \leq \|h_{i,j}\| \leq 1$ for every term. Moreover:

(a) **Spectral gap:**

- i. Every term $h_{i,j}^{(\text{snap})}$ in $H_2^{(\text{snap})}$ is η_2 -gapped on both subsystems i and j according to Definition 4.2.
- ii. Every term $h_{i,j}^{(\text{snap})}$ in $H_1^{(\text{snap})}$ is η_1 -gapped (according to Definition 4.2) on the subsystem acted upon nontrivially.
- iii. Every term $h_{i,j}^{(\text{snap})}$ in $H_0^{(\text{snap})}$ is proportional to the identity (zero spectral gap).

(b) **Rounding error:**

- i. Every term $h_{i,j}^{(\text{snap})}$ in $H_2^{(\text{snap})}$ is identical to the corresponding term $h_{i,j}$ in H .
- ii. For every term $h_{i,j}^{(\text{snap})}$ in $H_1^{(\text{snap})}$, we have

$$\|h_{i,j}^{(\text{snap})} - h_{i,j}\| \leq 4\eta_2. \quad (100)$$

This is because every term in $H_1^{(\text{snap})}$ was obtained by one application of snapping with gap parameter η_2 .

iii. For every term $h_{i,j}^{(\text{snap})}$ in $H_0^{(\text{snap})}$, we have

$$\|h_{i,j}^{(\text{snap})} - h_{i,j}\| \leq 4\eta_2 + 4\eta_1.$$

This is because every such term was obtained from snapping an original two-local term twice, first with gap parameter η_2 and then with gap parameter η_1 .

- (c) **Commutativity error:** Let $\varepsilon_1 := \varepsilon$. Write $\mathcal{S}_s(X) = \frac{1}{2}I_s \otimes \text{Tr}_s(X)$ for snapping qubit s , with tensor factors ordered appropriately. As shown in Theorem 4.5, this map is an average of Pauli conjugations and is therefore contractive in operator norm. If Y acts trivially on s , then

$$[\mathcal{S}_s(X), Y] = \mathcal{S}_s([X, Y]).$$

Recall that the sets of indices are distinct, so two sets of indices share at most one qubit. If snapping removes that shared qubit from either term, the resulting terms act on non-overlapping qubits and commute exactly. Otherwise, snapping acts only on qubits private to each term, so the identity above and contractivity apply. In particular:

- i. $H_2^{(\text{snap})}$ contains 2-local terms such that each pair of terms ε_2 -commute.
 - ii. $H_1^{(\text{snap})}$ contains 1-local terms such that each pair of terms ε_1 -commute.
- For two such terms overlapping on qubit i , write $h_{i,j}^{(\text{snap})} = \mathcal{S}_j(h_{i,j})$ and $h_{i,k}^{(\text{snap})} = \mathcal{S}_k(h_{i,k})$, where $j \neq k$. Then

$$\|[h_{i,j}^{(\text{snap})}, h_{i,k}^{(\text{snap})}]\| = \|\mathcal{S}_j\mathcal{S}_k([h_{i,j}, h_{i,k}])\| \leq \varepsilon = \varepsilon_1.$$

- iii. Any term in $H_2^{(\text{snap})}$ ε_1 -commutes with any term in $H_1^{(\text{snap})}$. Their commutator is zero unless they overlap on the qubit affected by the 1-local term. For $h_{i,j}^{(\text{snap})} = h_{i,j} \in H_2^{(\text{snap})}$ and $h_{i,k}^{(\text{snap})} = \mathcal{S}_k(h_{i,k}) \in H_1^{(\text{snap})}$, we have

$$\|[h_{i,k}^{(\text{snap})}, h_{i,j}^{(\text{snap})}]\| = \|\mathcal{S}_k([h_{i,k}, h_{i,j}])\| \leq \varepsilon = \varepsilon_1.$$

Terms in $H_0^{(\text{snap})}$ are scalar and commute with every term.

6. Pivot selection. For each qubit $i \in [n]$, assign it a pivot operator R_i as follows.

- (a) If qubit i is affected by at most one non-identity Hamiltonian term in $H^{(\text{snap})}$, let R_i be the one-qubit identity operator.
- (b) Otherwise, there are 2 subcases:
 - i. At least one of the terms $h_{i,j}^{(\text{snap})}$ in $H_1^{(\text{snap})}$ acts nontrivially on i . This term is 1-local: $h_{i,j}^{(\text{snap})} = g_i \otimes I_j$ for some g_i . In this case, set $R_i = g_i$.
 - ii. All of the terms in $H^{(\text{snap})}$ acting nontrivially on i are 2-local: in this case, pick an arbitrary one of these terms $h_{ij}^{(\text{snap})}$ (e.g. the one with the lexicographically first index pair i, j). By the assumption that $h_{ij}^{(\text{snap})} \in H_2^{(\text{snap})}$, it must be η_2 -gapped on both systems i and j . So writing its Pauli decomposition as

$$h_{ij}^{(\text{snap})} = \sum_{\alpha=0}^3 A_i^\alpha \otimes \sigma_j^\alpha,$$

we let R_i be any of the operators A_i^α that has spectral gap at least η_2 .

We claim that for each qubit $i \in [n]$ there exists a choice of parameters κ_i, ζ_i such that the following hold.

1. For any Hamiltonian term $h_{i,j}^{(\text{snap})} \in H^{(\text{snap})}$,

$$\|[h_{i,j}^{(\text{snap})}, R_i \otimes I_j]\| \leq \kappa_i,$$

and likewise with i and j interchanged.

2. For each i , either R_i is identity or the spectral gap of R_i is at least ζ_i .
3. For each i , $\kappa_i/\zeta_i \leq \max\{\varepsilon_1/\eta_1, 24\varepsilon_2/\eta_2^2\}$.

Case 1: at most 1 term in $H^{(\text{snap})}$ acts nontrivially on i . In this case, the algorithm above sets $R_i = I$. Let $\kappa_i = 0$ and $\zeta_i = 1$. The three properties hold.

Case 2: at least 2 distinct terms in $H^{(\text{snap})}$ act nontrivially on i . Within this case, there are two subcases.

Case 2a: at least one of the terms in $H^{(\text{snap})}$ acting nontrivially on i is 1-local. In this case, the procedure above sets R_i to be equal to (the nontrivial tensor factor of) some such 1-local term $h_{i,j}^{(\text{snap})}$. Let κ_i be equal to ε_1 and $\zeta_i = \eta_1$. We know that the term $h_{i,j}^{(\text{snap})}$ has commutativity error ε_1 with all other terms in $H^{(\text{snap})}$ and has spectral gap at least η_1 , so the three properties hold.

Case 2b: all of the terms in $H^{(\text{snap})}$ acting nontrivially on i are 2-local. In this case, the procedure above will take R_i to be some operator from the local decomposition of a term $h_{i,k}^{(\text{snap})}$. We will set $\kappa_i = 24\varepsilon_2/\eta_2$ and $\zeta_i = \eta_2$. This setting automatically satisfies the third property.

To see that the properties hold with these parameters, let the decomposition of $h_{i,k}^{(\text{snap})}$ be

$$h_{i,k}^{(\text{snap})} = \sum_{\beta=0}^3 B_i^\beta \otimes \sigma_k^\beta. \quad (101)$$

Let β^* be such that $R_i = B_i^{\beta^*}$. By construction, we know that β^* was chosen so that the spectral gap of $R_i = B_i^{\beta^*}$ is at least η_2 , which establishes the second property.

We now show the first property. Let $h_{i,j}^{(\text{snap})}$ be a Hamiltonian term acting on qubit i . Now, suppose $j \neq k$. Then decompose $h_{i,j}^{(\text{snap})}$ as

$$h_{i,j}^{(\text{snap})} = \sum_{\alpha=0}^3 A_i^\alpha \otimes \sigma_j^\alpha. \quad (102)$$

Since $\|[h_{i,j}^{(\text{snap})}, h_{i,k}^{(\text{snap})}]\| \leq \varepsilon_2$, we can conclude by applying Theorem 4.4 that for each α :

$$\|[A_i^\alpha, R_i]\| \leq \varepsilon_2, \quad (103)$$

and therefore

$$\|[h_{i,j}^{(\text{snap})}, R_i \otimes I_j]\| \leq 4\varepsilon_2 \leq 24\varepsilon_2/\eta_2, \quad (104)$$

under the assumption that $\eta_2 \leq 6$.

Now, suppose $j = k$. In this case, we are going to use transitivity (Lemma 3.6) to obtain our conclusion. Pick $j' \neq k$ such that $h_{i,j'}^{(\text{snap})}$ acts nontrivially on qubit i . (We are guaranteed that such a j' exists by the assumption that we are in Case 2.) Such Hamiltonian term has the decomposition

$$h_{i,j'}^{(\text{snap})} = \sum_{\alpha=0}^3 A_i^\alpha \otimes \sigma_{j'}^\alpha. \quad (105)$$

By Equation (103), we know that for each α , $\|[A_i^\alpha, R_i]\| \leq \varepsilon_2$. Now, let B_i^β be an operator from the decomposition of $h_{i,k}^{(\text{snap})}$. Since $\|[h_{i,k}^{(\text{snap})}, h_{i,j'}^{(\text{snap})}]\| \leq \varepsilon_2$, we can conclude by Theorem 4.4 that for any α :

$$\|[B_i^\beta, A_i^\alpha]\| \leq \varepsilon_2. \quad (106)$$

Furthermore, by the fact that we chose $h_{i,j'}$ to act nontrivially on qubit i , we know that it is η_2 -gapped on qubit i , and therefore that there exists an α^* such that $\Delta(A_i^{\alpha^*}) \geq \eta_2$.

Now, we will use transitivity: we know that $\|[B_i^\beta, A_i^{\alpha^*}]\| \leq \varepsilon_2$ and $\|[A_i^{\alpha^*}, R_i]\| \leq \varepsilon_2$, so by Lemma 3.6, we can conclude that

$$\|[B_i^\beta, R_i]\| \leq \frac{6\varepsilon_2}{\Delta(A_i^{\alpha^*})} \leq \frac{6\varepsilon_2}{\eta_2}, \quad (107)$$

where we are assuming that $\varepsilon_2 \leq \eta_2$ and moreover that the terms of the Hamiltonian have operator norm at most 1. So, we have

$$\|[h_{i,k}^{(\text{snap})}, R_i \otimes I_k]\| \leq \frac{24\varepsilon_2}{\eta_2}.$$

7. **Pinch everything.** Define the global pinching map $\mathcal{P}$ that pinches each qubit i by R_i . Our final Hamiltonian will be

$$\hat{H} = \mathcal{P}(H^{(\text{snap})}).$$

Tensor-product pinching is an average of unitary conjugations and is therefore contractive in operator norm. Consequently, $\|\hat{h}_{i,j}\| \leq \|h_{i,j}^{(\text{snap})}\| \leq \|h_{i,j}\| \leq 1$. By Lemma 5.2, we conclude that the resulting Hamiltonian $\hat{H}$ is commuting, and that for each term $\hat{h}_{i,j}$ in $\hat{H}$, its distance from the corresponding term $h_{i,j}^{(\text{snap})}$ in $H^{(\text{snap})}$ is bounded by

$$\|\hat{h}_{i,j} - h_{i,j}^{(\text{snap})}\| \leq 4 \max_{i,j} \left\{ \left(\frac{\kappa_i}{\zeta_i} + \frac{\kappa_j}{\zeta_j} \right) \right\} \leq 8 \max \left\{ \frac{\varepsilon_1}{\eta_1}, \frac{24\varepsilon_2}{\eta_2^2} \right\} \leq 192 \varepsilon^{1/3},$$

where the final inequality follows from the parameter settings we have made:

$$\varepsilon_2 = \varepsilon, \quad (108)$$

$$\varepsilon_1 = \varepsilon, \quad (109)$$

$$\eta_2 = \sqrt[3]{\varepsilon}, \text{ and} \quad (110)$$

$$\eta_1 = \eta_2 = \sqrt[3]{\varepsilon}. \quad (111)$$

Computing the total distance between the input and output terms for each term. For each input Hamiltonian term $h_{i,j}$, we compute an upper bound on its distance from the output term $\hat{h}_{i,j}$. By the triangle inequality, there are two distances to add: (1) the distance $\|h_{i,j}^{(\text{snap})} - \hat{h}_{i,j}\|$, which we showed above is bounded by $192 \varepsilon^{1/3}$, and (2) the distance $\|h_{i,j} - h_{i,j}^{(\text{snap})}\|$, which we bounded earlier by $4(\eta_2 + \eta_1)$. Thus, the total distance is bounded above as follows.

$$\begin{aligned} \|\hat{h}_{i,j} - h_{i,j}\| &\leq \|\hat{h}_{i,j} - h_{i,j}^{(\text{snap})}\| + \|h_{i,j}^{(\text{snap})} - h_{i,j}\| \\ &\leq 192 \varepsilon^{1/3} + 4\eta_2 + 4\eta_1 \\ &\leq 192 \varepsilon^{1/3} + 4\sqrt[3]{\varepsilon} + 4\varepsilon^{1/3} && \text{Ineq. 110 and Ineq. 111} \\ &\leq 200 \varepsilon^{1/3}. \end{aligned}$$

From this, the total operator norm bound

$$\|\hat{H} - H\| \leq 200m\varepsilon^{1/3}$$

follows directly by applying the triangle inequality m times.

Runtime: The fact that the runtime is linear in m follows by inspecting the algorithm: only the at most $2m$ qubits occurring in the input need to be processed, and all matrix operations are on constant-size matrices. For rational input entries and rational ε , the gap comparisons involve algebraic numbers of bounded degree and polynomial bit complexity. The output terms can themselves be represented exactly with rational entries: midpoint snapping is rational, and for a non-scalar rational pivot R , writing $B = R - \frac{\text{Tr}(R)}{2}I$ and $b^2 = \text{Tr}(B^2)/2 > 0$ gives

$$\mathcal{P}_R(X) = \frac{1}{2} \left(X + \frac{BXB}{b^2} \right).$$

Each output term uses only a constant number of these rational matrix operations, so its bit complexity is polynomial in that of the input. In particular, exact commutation does not require rounding the output matrices to floating-point numbers. $\square$

Lemma 5.2 (The multi-qubit pinching lemma). *Let $H = \sum_{(i,j) \in E} h_{ij}$ be a 2-local Hamiltonian and for each $i \in [n]$, let $\mathcal{P}_i$ be either the identity superoperator or the pinching superoperator for some one-local Hermitian operator R_i acting on qubit i . Suppose that for each i where $\mathcal{P}_i$ is not identity, $\Delta(R_i) \geq \eta_i$ and for all j ,*

$$\|[R_i \otimes I_j, h_{i,j}]\| \leq \kappa_i,$$

for parameters $\eta_i > 0$ and $\kappa_i \geq 0$; when $\mathcal{P}_i = \text{id}$, set $\eta_i = 1$ and $\kappa_i = 0$. Then the 2-local Hamiltonian

$$H' = \sum_{(i,j) \in E} (\mathcal{P}_i \otimes \mathcal{P}_j \otimes \text{id}_{[n] \setminus \{i,j\}})(h_{i,j}) \quad (112)$$

satisfies the following properties:

1. *For all i such that $\mathcal{P}_i$ is not identity, all the terms $h'_{i,j}$ touching qubit i commute with each other, and with R_i .*
2. *For all $(i,j) \in E$,*

$$\|h'_{i,j} - h_{i,j}\| \leq 4 \left(\frac{\kappa_i}{\eta_i} + \frac{\kappa_j}{\eta_j} \right). \quad (113)$$

Proof. We show the two properties in turn.

1. **Commutativity.** We only need to check commutativity for pairs of terms which overlap on exactly one qubit (there are no two terms that overlap on two qubits since our convention¹⁴ is that $h_{i,j}$ is the sole term in the Hamiltonian acting on the pair i, j), and that were affected nontrivially by the pinching on the shared qubit. Let $\{i, j, k\}$ be three distinct indices. Consider the terms $h_{i,j}$ and $h_{j,k}$ which have the following decomposition according to Lemma 4.1.

$$h_{i,j} = \sum_{\alpha=0}^3 \sigma_i^\alpha \otimes A_j^\alpha \quad h_{j,k} = \sum_{\beta=0}^3 B_j^\beta \otimes \sigma_k^\beta \quad (114)$$

Now, consider the commutator $[h'_{i,j}, h'_{j,k}]$.

$$\begin{aligned} [h'_{i,j}, h'_{j,k}] &= [(\mathcal{P}_i \otimes \mathcal{P}_j)(h_{i,j}), (\mathcal{P}_j \otimes \mathcal{P}_k)(h_{j,k})] \quad (\text{by Equation (112)}) \\ &= \left[\sum_{\alpha} (\mathcal{P}_i \otimes \mathcal{P}_j)(\sigma_i^\alpha \otimes A_j^\alpha), \sum_{\beta} (\mathcal{P}_j \otimes \mathcal{P}_k)(B_j^\beta \otimes \sigma_k^\beta) \right] \quad (\text{by Equation (114)}) \\ &= \sum_{\alpha, \beta} \mathcal{P}_i(\sigma_i^\alpha) \otimes \underbrace{[\mathcal{P}_j(A_j^\alpha), \mathcal{P}_j(B_j^\beta)]}_{=0} \otimes \mathcal{P}_k(\sigma_k^\beta) \quad (\text{commutator properties}) \\ &= 0. \end{aligned}$$

To see that the commutators in the sum are 0, observe that by Lemma 3.2, each $\mathcal{P}_j(A_j^\alpha)$ and $\mathcal{P}_j(B_j^\beta)$ commutes with R_j , and thus by Lemma 3.5 and the fact that R_j is gapped, we get that $\mathcal{P}_j(A_j^\alpha)$ and $\mathcal{P}_j(B_j^\beta)$ commute with each other. The same calculation shows that each $h'_{i,j}$ commutes with R_i whenever $\mathcal{P}_i$ is not identity.

2. **Rounding error.** To bound the rounding error, we will first show a commutativity property of the operators R_i for every qubit where they are defined. Consider such a qubit i and let $h_{i,j}$ be a Hamiltonian term acting on qubit i . Decompose $h_{i,j}$ as

$$h_{i,j} = \sum_{\alpha=0}^3 A_i^\alpha \otimes \sigma_j^\alpha. \quad (114)$$

Since, by the hypothesis, $\|[h_{i,j}, (R_i \otimes I_j)]\| \leq \kappa_i$, expanding this commutator and applying Lemma 4.3 shows that for each α :

$$\|[A_i^\alpha, R_i]\| \leq \kappa_i. \quad (115)$$

Now, we can compute the rounding error of any Hamiltonian term. Let $h_{i,j}$ be any term of the Hamiltonian. If both $\mathcal{P}_i$ and $\mathcal{P}_j$ are equal to identity, then there is nothing to prove.

Now, suppose without loss of generality that qubit i has $\mathcal{P}_i$ not equal to identity, so

¹⁴This convention is without loss of generality, since one can absorb all terms that act on a pair i, j into a single term at the start of the analysis.

R_i is defined. Then we obtain

$$\|\mathcal{P}_i(h_{i,j}) - h_{i,j}\| = \left\| \sum_{\alpha=0}^3 (\mathcal{P}_i(A_i^\alpha) - A_i^\alpha) \otimes \sigma_j^\alpha \right\| \quad (116)$$

$$\leq \sum_{\alpha=0}^3 \|\mathcal{P}_i(A_i^\alpha) - A_i^\alpha\| \quad (117)$$

$$\leq \frac{1}{\Delta(R_i)} \sum_{\alpha=0}^3 \|[A_i^\alpha, R_i]\| \quad (118)$$

$$\leq \frac{4}{\Delta(R_i)} \cdot \kappa_i \leq \frac{4\kappa_i}{\eta_i}, \quad (119)$$

where in the second to last line we applied Lemma 3.2 and Equation (115). Now, if $\mathcal{P}_j$ is equal to identity, then $h'_{i,j} = \mathcal{P}_i(h_{i,j})$, so we have shown Equation (113) in this case.

The remaining case is if $\mathcal{P}_j$ is also not equal to identity. In this case, write the decomposition

$$h_{i,j} = \sum_{\alpha=0}^3 \sigma_i^\alpha \otimes C_j^\alpha.$$

We can similarly derive the bound

$$\|(\mathcal{P}_i \otimes \mathcal{P}_j)(h_{i,j}) - \mathcal{P}_i(h_{i,j})\| = \left\| \sum_{\alpha=0}^3 \mathcal{P}_i(\sigma_i^\alpha) \otimes (\mathcal{P}_j(C_j^\alpha) - C_j^\alpha) \right\| \quad (120)$$

$$\leq \sum_{\alpha=0}^3 \|(\mathcal{P}_j(C_j^\alpha) - C_j^\alpha)\| \quad (121)$$

$$\leq \frac{1}{\Delta(R_j)} \sum_{\alpha=0}^3 \|[C_j^\alpha, R_j]\| \quad (122)$$

$$\leq \frac{4\kappa_j}{\eta_j} \quad (\text{by Equation (115) and } \Delta(R_j) \geq \eta_j), \quad (123)$$

where we have used the bound $\|\mathcal{P}_i(\sigma_i^\alpha)\| \leq \|\sigma_i^\alpha\| = 1$, which follows from the fact that $\mathcal{P}_i$ is a positive unital map and thus contractive in operator norm (see [Wat18, Theorem 6.12]). So overall, by the triangle inequality, we get

$$\|h_{i,j} - (\mathcal{P}_i \otimes \mathcal{P}_j)(h_{i,j})\| \leq \|h_{i,j} - \mathcal{P}_i(h_{i,j})\| + \|\mathcal{P}_i(h_{i,j}) - (\mathcal{P}_i \otimes \mathcal{P}_j)(h_{i,j})\| \quad (124)$$

$$\leq \frac{4\kappa_i}{\eta_i} + \frac{4\kappa_j}{\eta_j}. \quad (125)$$

This proves Equation (113). □

6 Applications to Almost Commuting 2-local Hamiltonians

In this section, we use our locality-preserving rounding technique (formally, Theorem 5.1) to show that the almost commuting 2-local Hamiltonian problem lies in NP.

6.1 Containment in NP

Theorem 6.1. *There exists a linear-time computable reduction from the (γ, ε) -2-ACLH problem to the (γ') -2-CLH where $\gamma' = \gamma - 400 \varepsilon^{1/3}$ for any $0 \leq \varepsilon \leq 1$ such that $\gamma' \geq 1/\text{poly}(n)$.*

Proof. Given (H, a, b) an instance of (γ, ε) -2-ACLH, let H' be the commuting Hamiltonian obtained by applying Theorem 5.1 to H , and let $a' = a + 200m\varepsilon^{1/3}$ and $b' = b - 200m\varepsilon^{1/3}$. These thresholds have exact polynomial-bit descriptions as constant-degree algebraic numbers (a defining polynomial together with an isolating interval); their numerical approximations can be computed to any inverse-polynomial accuracy in polynomial time. From the operator norm bound in the theorem, together with Fact 2.4, we know that

$$|\lambda_{\min}(H') - \lambda_{\min}(H)| \leq 200m\varepsilon^{1/3}.$$

Hence, we have

$$\lambda_{\min}(H) \leq a \implies \lambda_{\min}(H') \leq a' \tag{126}$$

$$\lambda_{\min}(H) \geq b \implies \lambda_{\min}(H') \geq b'. \tag{127}$$

Thus, since $(b' - a')/m = \gamma' \geq 1/\text{poly}(n)$, it holds that (H', a', b') is always an instance of (γ') -2-CLH satisfying the promise for $\gamma' = \gamma - 400\varepsilon^{1/3}$, and H' is a YES instance iff H is a YES instance, so the map $H \mapsto H'$ is a reduction as claimed. $\square$

Corollary 6.2. *The (γ, ε) -2-ACLH promise problem is in NP when $\gamma - 400\varepsilon^{1/3} \geq 1/\text{poly}(n)$.*

Proof. This is a direct consequence of Theorem 6.1 and Theorem 2.15. $\square$

6.2 Quantum Gibbs sampling

As an application of our locality-preserving rounding technique for almost commuting quantum many-body Hamiltonians, we now consider the task of quantum Gibbs sampling. For a local Hamiltonian H and an inverse temperature $\beta > 0$, the Gibbs state

$$\rho_{\beta}(H) = \frac{e^{-\beta H}}{\text{Tr}[e^{-\beta H}]} \tag{128}$$

describes the thermal equilibrium properties of a quantum system at finite temperature $T = 1/\beta$. Given a precision parameter $\delta > 0$, the (H, β, δ) -Gibbs state preparation (or sampling) task with respect to the Hamiltonian H and an inverse temperature $\beta > 0$ is to output ρ such that

$$\|\rho - \rho_{\beta}(H)\|_1 \leq \delta. \tag{129}$$

Almost-commuting Hamiltonians. Our rounding technique allows us to reduce quantum Gibbs sampling for ε -almost-commuting 2-local qubit Hamiltonians to the task of Gibbs sampling for a nearby commuting 2-local qubit Hamiltonian, provided that ε is sufficiently small.

We use the following result on the continuity of the Gibbs state (with respect to the Hamiltonian) which is implicit in [CMTW25, Proposition 15].

Proposition 1 (Continuity of the Gibbs state, [CMTW25]). *Let $H, \hat{H}$ be Hermitian operators acting on a finite-dimensional complex vector space. Then, for any inverse temperature $\beta > 0$,*

$$\|\rho_\beta(H) - \rho_\beta(\hat{H})\|_1 \leq e^{2\beta\|H - \hat{H}\|} - 1.$$

In particular, if $\|H - \hat{H}\| \leq \ln(1 + \delta)/2\beta$, then it holds that $\|\rho_\beta(H) - \rho_\beta(\hat{H})\|_1 \leq \delta$.

Equipped with the above technical fact, we can now show the following reduction.

Theorem 6.3. *Let $H = \sum_{i=1}^m h_i$ be an ε -almost-commuting 2-local qubit Hamiltonian with $\|h_i\| \leq 1$ and $\|[h_i, h_j]\| \leq \varepsilon$, for all i, j , where $0 \leq \varepsilon \leq 1$. Then, for any inverse temperature $\beta > 0$ and any precision $\delta > 0$ such that*

$$200m\varepsilon^{1/3} \leq \ln\left(1 + \frac{\delta}{2}\right)/2\beta \quad (130)$$

there exists a reduction from the task of (H, β, δ) -Gibbs sampling to the task of $(\hat{H}, \beta, \delta/2)$ -Gibbs sampling for a 2-local commuting qubit Hamiltonian $\hat{H}$. Moreover, if the Gibbs sampler for $\hat{H}$ runs in time T , then the resulting Gibbs sampler for H runs in time $T + O(m)$.

Proof. The reduction simply applies the rounding technique from Theorem 5.1 to H , resulting in a 2-local commuting qubit Hamiltonian $\hat{H}$ such that $\|H - \hat{H}\| \leq 200m\varepsilon^{1/3}$. Then, the output of any successful $(\hat{H}, \beta, \delta/2)$ -Gibbs sampler results in a state ρ such that

$$\begin{aligned} \|\rho - \rho_\beta(H)\|_1 &\leq \|\rho - \rho_\beta(\hat{H})\|_1 + \|\rho_\beta(\hat{H}) - \rho_\beta(H)\|_1 && \text{(triangle inequality)} \\ &\leq \frac{\delta}{2} + \|\rho_\beta(\hat{H}) - \rho_\beta(H)\|_1 && \text{(by assumption)} \\ &\leq \frac{\delta}{2} + e^{2\beta\|H - \hat{H}\|} - 1 && \text{(Proposition 1)} \\ &\leq \frac{\delta}{2} + \frac{\delta}{2} = \delta && \text{(due to (130))} \end{aligned}$$

Because the rounding step is deterministic and requires time $O(m)$, the claim follows. $\square$

We remark that our Gibbs sampling reduction for almost-commuting Hamiltonians above can be further combined with the reduction of [HJ25, Theorem 9], thereby reducing Gibbs sampling for (certain) almost-commuting 2-local qubit Hamiltonians to Gibbs sampling of certain classical Hamiltonians.

6.3 Faster Hamiltonian simulation

Our result gives a method to speed up Hamiltonian simulation for approximately commuting Hamiltonians. The basic idea is to use the fact that time evolution of commuting Hamiltonians can be simulated very efficiently, since the matrix exponential factors into a product of exponentials of the individual terms. For short times, thus, one could simply round an approximately commuting Hamiltonian H to a commuting Hamiltonian H' with our technique, and simulate H' instead of H . For longer times, the error incurred by this rounding may be prohibitive: for this more general case, we can use the "interaction picture" formalism of Low and Wiebe [LW18], in essentially a black-box way, to add corrections from the rounding error.

As a first step, we show the following helper lemma which is essentially folklore.

Lemma 6.4. Suppose $H = \sum_{i=1}^m h_i$ is an $O(1)$ -local commuting Hamiltonian on n qubits, with $\|h_i\| \leq 1$, $m = \Omega(n)$, $t \in \mathbb{R}$, and $0 < \delta < 1$. Then there exists a unitary quantum circuit on n system qubits and one ancillary qubit of gate complexity

$$O\left(m \text{poly}(\log(m/\delta), \log(2 + |t|))\right)$$

that implements a unitary U such that, with the ancillary qubit initialized to $|0\rangle$,

$$\|U(I \otimes |0\rangle) - (e^{iHt} \otimes |0\rangle)\| \leq \delta.$$

Moreover, this circuit can be computed from a classical description of H and t in classical time scaling as $O(m \text{poly}(L, \log(m/\delta), \log(2 + |t|)))$, where L bounds the bit complexity of each local term and of t .

Proof. Using the commutativity of the terms of H , we may write the exponential of H as a product of exponentials of individual terms as follows:

$$e^{iHt} = e^{i \sum_{i=1}^m h_i t} = \prod_{i=1}^m e^{ih_i t}.$$

So to simulate the time evolution, it suffices to simulate $e^{ih_i t}$ for each term h_i . To do this, since h_i acts locally on a constant number of qubits, we can explicitly compute $e^{ih_i t}$, up to some appropriate finite precision, and then apply the Solovay-Kitaev theorem, in the algorithmic form of [DN06], to simulate it using a sequence of elementary gates. To retain scalar phases, embed each local unitary V as $V \otimes |0\rangle\langle 0| + V^\dagger \otimes |1\rangle\langle 1|$ on one additional qubit. This embedded unitary has determinant one and acts as V when the ancillary qubit is $|0\rangle$. The same ancillary qubit can be reused throughout, and the approximation errors accumulate by the triangle inequality.

We now estimate the time complexity of this procedure. In order to implement e^{iHt} up to error δ , we need to implement each $e^{ih_i t}$ up to error δ/m . To do this, if each term h_i is k -local, it suffices to compute each entry of the matrix $e^{ih_i t}$ up to error $\delta/(100m2^k)$ (so that the total computed matrix is at most $\delta/(100m)$ far in operator norm from $e^{ih_i t}$). Replacing this numerical matrix by its unitary polar factor increases the error by at most a constant factor. We then apply the Solovay-Kitaev theorem to approximate its determinant-one embedding up to error $\delta/(100m)$. Applying the results of [DN06], we get the claimed gate complexity and runtime. $\square$

We will now sketch how to use our rounding to perform Hamiltonian simulation. The main technical obstacle to directly applying a standard algorithm is the following: Hamiltonian simulation algorithms are typically given in the oracle model, where access to H is given through a unitary oracle O_H that implements a "block-encoding" of H : $(\langle 0| \otimes I)O_H(|0\rangle \otimes I) \propto H$. However, in our setting, we need to work in a model where the Hamiltonian is given as *explicit* input (i.e. as a classical string containing descriptions of all the local terms), in order to apply our rounding algorithm. Indeed, implicit in the oracle picture is the assumption that for any realistic Hamiltonian, such an oracle can be implemented efficiently. So we will assume that we have a method to construct such a block encoding and leave its size as a parameter.

Our rounding says that

$$H = H' + \Delta,$$

where H' is a commuting Hamiltonian with $\|H'\| \leq \alpha_A := m$, and Δ is a local Hamiltonian with $\|\Delta\| \leq 200m\varepsilon^{1/3}$. Here we assume the hypotheses of Theorem 5.1. Moreover, H' and

Δ are efficiently computable from H . If $\Delta = 0$, we use Lemma 6.4 alone. Otherwise, choose a known normalization $\alpha_B > 0$ such that

$$\|\Delta\| \leq \alpha_B \leq 200m\varepsilon^{1/3};$$

for example, one may take $\alpha_B = 200m\varepsilon^{1/3}$. Suppose that, once we have written down Δ , we can construct a block encoding of Δ/α_B , together with its controlled version and inverse, with gate complexity T_{block} at the precision required below. We assume that the number of block-encoding ancillas is $O(T_{block})$, so reflections on those ancillas are included in the cost bound.

We use the interaction-picture construction of Low and Wiebe [LW18], with H' as the “fast” part and Δ as the slow part. To account for small times explicitly, let $t > 0$, $0 < \delta < 1$, and set

$$r = \max\{1, \lceil 2\alpha_B t \rceil\}, \quad \tau = t/r, \quad H_I(s) = e^{iH's} \Delta e^{-iH's} \quad (0 \leq s \leq \tau).$$

Then

$$e^{-iHt} = \left(e^{-iH'\tau} \mathcal{T} e^{-i \int_0^\tau H_I(s) ds} \right)^r, \quad \|H_I(s)\| \leq \alpha_B, \quad \|\dot{H}_I(s)\| \leq 2\alpha_A \alpha_B.$$

The truncated-Dyson-series algorithm of [LW18, Theorem 3 and Lemma 6], with error $O(\delta/r)$ per segment, uses $\tilde{O}(r)$ queries in total to a time-indexed block encoding of $H_I(s)/\alpha_B$, together with r evolutions under H' .

For completeness, the required time-indexed oracle can be implemented directly, without assuming a simulation cost proportional to the duration of each short evolution. As in the proof of [LW18, Theorem 7], let

$$V = \sum_{\ell=0}^{M-1} |\ell\rangle\langle\ell| \otimes I_a \otimes e^{-iH'\ell\tau/M},$$

where I_a acts on the block-encoding ancillas. If O_Δ block-encodes Δ/α_B , then $V^\dagger(I \otimes O_\Delta)V$ is the required time-indexed block encoding. Here M can be chosen to be a power of two of order $1 + t(\alpha_A + \alpha_B)/\delta + \log^2(2 + r/\delta)$, as required by [LW18, Lemma 5]. Expanding ℓ in binary implements this controlled evolution using $O(\log M)$ controlled evolutions under H' , each of duration at most $\tau \leq t$. The proof of Lemma 6.4 applies to these controlled evolutions as well: controlling each local exponential increases its locality by only one. Each such evolution therefore costs $\tilde{O}(m)$ gates, uniformly in its duration, when its error is chosen sufficiently small that all oracle-synthesis errors sum to $O(\delta)$. Thus a time-indexed oracle costs $\tilde{O}(T_{block} + m)$ gates. The r separate evolutions under H' and the auxiliary gates in the Dyson-series algorithm fit within the same total bound.

Putting this together and ignoring polylogarithmic factors gives gate complexity

$$\tilde{O}((1 + \alpha_B t)(T_{block} + m)), \quad \text{and hence} \quad \tilde{O}((1 + m\varepsilon^{1/3}t)(T_{block} + m)),$$

where the hidden factors are polynomial in $\log(2 + m(1 + t)/\delta)$. The case $t = 0$ requires no gates, and negative times are handled by taking the inverse circuit. Apart from the commuting-simulation overhead and polylogarithmic factors, the important feature is that the time-dependent cost scales with a normalization bound for Δ , controlled by the commutator error, rather than with the total norm of the Hamiltonian.

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A Lower bound on the term-wise distance

We now show a concrete pair of 2-qubit hermitian operators that ε -almost commute for which any exactly commuting approximation needs to move one of them by $\Omega(\sqrt{\varepsilon})$.

Theorem A.1 (Lower bound of rounding two anticommuting binary observables). *Let P, Q be a pair of anti-commuting ($PQ = -QP$) binary observables on a Hilbert space $\mathcal{H}$. Let $t > 0$. Then,*

1. $\varepsilon := \|[tP, tQ]\| = 2t^2$, and

2. for any pair of commuting Hermitian operators $\hat{P}, \hat{Q}$, the following holds:

$$\|tP - \hat{P}\|^2 + \|tQ - \hat{Q}\|^2 \geq t^2 = \varepsilon/2. \quad (131)$$

In other words, in any rounding scheme, one of the rounded operators must be moved by a distance of at least $\frac{\sqrt{\varepsilon}}{2}$.

Proof. Since $\hat{P}$ and $\hat{Q}$ commute, they are simultaneously diagonalizable, and thus have a common eigenvector $|\psi\rangle$. Thus,

$$\hat{P}|\psi\rangle = \hat{p}|\psi\rangle, \quad \hat{Q}|\psi\rangle = \hat{q}|\psi\rangle. \quad (132)$$

Also, define p and q as follows:

$$p := \langle\psi|P|\psi\rangle \quad \text{and} \quad q := \langle\psi|Q|\psi\rangle \quad (133)$$

where p and q are real since P and Q are Hermitian. We now have:

$$\begin{aligned} \|tP - \hat{P}\|_{\text{op}}^2 &\geq \|(tP - \hat{P})|\psi\rangle\|_2^2 && \text{by Definition of } \|\cdot\|_{\text{op}} \\ &= \|tP|\psi\rangle - \hat{p}|\psi\rangle\|_2^2 \\ &= t^2 \langle\psi|P^2|\psi\rangle - 2t\hat{p}\langle\psi|P|\psi\rangle + \hat{p}^2 \langle\psi|\psi\rangle \\ &= t^2 - 2tp\hat{p} + \hat{p}^2 \\ &= (\hat{p} - tp)^2 + t^2(1 - p^2) \geq t^2(1 - p^2). \end{aligned}$$

Similarly, we have:

$$\|tQ - \hat{Q}\|^2 \geq t^2(1 - q^2). \quad (134)$$

Thus,

$$\|tP - \hat{P}\|^2 + \|tQ - \hat{Q}\|^2 \geq t^2(2 - (p^2 + q^2)). \quad (135)$$

If $p^2 + q^2 \leq 1$, we obtain our result. We show that this is the case during the rest of this proof. Define

$$R := pP + qQ \quad \text{and} \quad s := p^2 + q^2. \quad (136)$$

By anticommutation ($PQ + QP = 0$) and since $P^2 = Q^2 = I$, we have:

$$R^2 = (pP + qQ)^2 \quad (137)$$

$$= p^2 P^2 + pq(PQ + QP) + q^2 Q^2 \quad (138)$$

$$= p^2 \cdot I + 0 + q^2 \cdot I \quad (139)$$

$$= (p^2 + q^2) \cdot I = s \cdot I. \quad (140)$$

Since R is hermitian, $\|R\|^2 = \|R^2\| = |s| = s$ (since $s = p^2 + q^2$ is non-negative). Thus, $\|R\| = \sqrt{s}$. On the other hand, we have:

$$\langle\psi|R|\psi\rangle = \langle\psi|pP|\psi\rangle + \langle\psi|qQ|\psi\rangle = p^2 + q^2 = s. \quad (141)$$

By the definition of operator norm, we have $s \leq \|R\| = \sqrt{s}$. Thus, $s \leq 1$. $\square$

Corollary A.2. For $0 < \varepsilon \leq 1$, set

$$h_{12} = tX_1X_2, \quad h_{13} = tZ_1Z_3, \quad t = \sqrt{\varepsilon/2}, \quad (142)$$

and let $\hat{P}$ and $\hat{Q}$ be a pair of Hermitian commuting operators. Then, $\|[h_{12}, h_{13}]\| = \varepsilon$, but:

$$\|h_{12} - \hat{P}\| \geq \frac{\sqrt{\varepsilon}}{2} \quad \text{or} \quad \|h_{13} - \hat{Q}\| \geq \frac{\sqrt{\varepsilon}}{2}. \quad (143)$$

Proof. Since X_1 and Z_1 anti-commute, and since the Pauli matrices are Hermitian observables, the result follows directly by Theorem A.1. $\square$